

When Does KV-Cache Eviction Help versus Hurt? A Task-Type Partition and a Label-Free Gating Predictor

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Abstract

Recent work shows that KV-cache eviction can do more than compress: in long-context LLMs, selective retention can match or exceed full-cache accuracy because eviction removes the attention dilution produced by redundant prefill tokens. This raises a question those papers do not address: on which inputs and tasks does the effect actually occur, and can we tell in advance? We study six RULER task families at context lengths 4K and 16K across three architecture families (Qwen2.5 at 1.5B/3B/14B/32B; Mistral-7B-v0.3; and two Llama-family models, Yi-1.5-9B-Chat and Llama-3.1-8B-Instruct) and find a task-type partition. Five task families (multi-hop variable tracking, frequent-words extraction, single- and multi-document QA, multi-value retrieval) are *dilution-prone*: eviction recovers accuracy on a non-trivial input fraction. One task family (NIAH-MK3, precise multi-needle retrieval) is *capacity-bound*: under attention-score-based evictors, eviction is monotone damaging, with $\rho \leq 0.02$ in every 4K/16K cell and 0.04 in a 32K extension, $15\times$ below the dilution-task recovery ratio there. We introduce a label-free a priori predictor of which side an input falls on: the early-to-late head-agreement drop on prefill attention. A single threshold $\tau = 0.07$, calibrated once on Qwen2.5-1.5B, separates the partition on 7 of 8 Qwen/Mistral (model $\times$ context) cells; a 40-input recipe fixes the eighth. Transfer of the raw threshold to three further families (Yi-1.5, Llama-3.1, and Qwen3, whose drop scale differs enough that fixed τ never fires) is partial, but standardizing the drop per model on an unlabeled pilot (z-scoring) restores zero-shot transfer of a single global threshold. The drop powers a drop-in gating wrapper around any SnapKV-style evictor. On a 4×4 matrix of four base evictors (SnapKV, H2O, StreamingLLM, PyramidKV) and four model sizes / architectures, gating lifts every cell with grand-mean $+22.9$ pp at matched nominal budget (a $3.1\text{--}3.4\times$ achieved memory reduction at the most aggressive operating point), and even the best plain evictor is beaten by some gated evictor by $+13$ to $+29$ pp. On the strongest single-task showcase (Mistral-7B-v0.3, RULER 4K NIAH-MK3, $N = 100$), plain SnapKV collapses from 99% to 0% as budget shrinks, while gated SnapKV holds 89% at every budget. We supply a context-length scaling formula $\rho \approx (1 - A_{\text{full}}) \cdot p_{\text{recov}}(\rho_{\mathcal{R}}/\gamma)$ that simultaneously explains context amplification, model-size saturation, and the recovery of the dilution regime at long context in larger models.

1 Introduction

Long-context LLMs spend most of their inference cost on the KV-cache, and a sequence of recent papers shows that eviction can do more than compress. Bui et al. [2026] prove that selective retention can reduce a quantifiable *attention dilution* term and report a learned-gate evictor (DBTrimKV) that exceeds the full cache on long-context dialogue by several points. Yang et al. [2026a] derive a closed-form information-bottleneck score that produces the same paradox on LongBench; an indexer-and-latent-memory architecture [Yang et al., 2026c] reports an information-density peak on RULER and LongBench at moderate compression. The common claim across these papers is that less compute can yield more accuracy.

Open question. Each of these results reports the effect averaged over a benchmark suite where it occurs. None addresses two natural follow-up questions:

- (Q1) *For which inputs and tasks does eviction help, and for which does it hurt?* Mean accuracy hides per-input variance: a benchmark mean that is positive in expectation can mask a subset of inputs on which eviction is catastrophic.
- (Q2) *Can the partition be predicted a priori from cheap inference-time signals?* Existing methods that adapt either train a scorer (Bui et al., 2026, Yang et al., 2026c, Dong et al., 2026, Moschella et al., 2026) or derive an unconditional capacity score from query statistics (Yang et al., 2026a). Whether the dilution-prone vs. capacity-bound distinction is visible in the pre-eviction attention alone is open.

We answer Q1 and Q2 affirmatively. The *head-agreement drop* on prefill attention, the difference between mean Jaccard top- k head agreement in early vs. late layer-bins, partitions inputs into a dilution-prone class $\mathcal{D}$ and a capacity-bound class $\mathcal{C}$, and a drop-in gating wrapper turns the partition into a training-free policy that strictly improves any SnapKV-style evictor at matched budget.

Contributions.

1. **An empirical partition** (Section 3). Across eight (model, context) cells (Qwen2.5-1.5B/3B at 4K and 16K, Qwen2.5-14B at 4K and 16K, Mistral-7B-v0.3 at 4K and 16K) and six RULER task families, NIAH-MK3 is the only task with $\rho \leq 0.02$ in every cell; the five dilution-prone tasks have $\rho \geq 0.05$ at ≥ 1 context length unless the model saturates A_{full} (scaling-formula consistent) or the cell’s residual errors are budget-insensitive (three such cells, all QA: QA_2 on Qwen-14B, QA_1 and QA_2 on Mistral; Section 3). A controlled near-tie distractor sweep (niah_multikey_1/2/3) confirms the mechanism: D shrinks monotonically from 0.155 to 0.045 as distractor count grows from 0 to 2, and per-input D ranges between MK2 and MK3 do not overlap on Qwen2.5-1.5B. Yi-1.5-9B-Chat and Llama-3.1-8B enter as Llama-architecture cross-architecture tests with partial transfer (4 of 5 dilution-prone tasks match prediction; niah_multivalue mis-classifies on both Llama-family models; see Section 3).
2. **A label-free predictor** (Section 4.1). The early-to-late head-agreement drop $D := \text{agr}_{\text{early}} - \text{agr}_{\text{late}}$ is the smallest on NIAH-MK3 on Qwen and Mistral cells; on Yi-1.5 and Llama-3.1 it ranks 2nd-smallest behind niah_multivalue (Llama-arch family artifact). Cross-architecture transfer at $\tau = 0.07$ holds in 7 of 8 Qwen/Mistral (model, context) cells; Section 5 fixes the eighth (Qwen 3B 16K) with a 40-input calibration recipe, and a per-model z-scoring of the drop (unlabeled pilot, one global threshold fit on Qwen/Mistral only) transfers zero-shot to both Llama-family models. A τ sensitivity sweep (Figure 2) identifies a robustness plateau $\tau \in [0.055, 0.10]$ inside which Δ varies by under 1pp.
3. **A gating algorithm** (Section 4). One conditional on the prefill attention, no training, no per-input back-prop, no labels. On a 4×4 (base evictor $\times$ Qwen2.5/Mistral) matrix the grand-mean Δ is +22.9pp at matched budget. Best plain evictor beaten by best gated by +13 to +29pp across all four models. Cross-architecture SnapKV runs on Yi-1.5-9B ($\Delta = +39.0\text{pp}$) and Llama-3.1-8B ($\Delta = +10.9\text{pp}$) show the wrapper also helps on Llama-family models, though there part of the gain comes from conservative gate-closing rather than a correctly recovered partition (Section 8).
4. **Direct head-to-heads against published methods** (Sections 7.12, 7.13). Wrapping CapKV (Yang et al., 2026a) in our gate on LongBench-qasper and LongBench-triviaqa yields $\Delta = +1.5\text{pp}$ and $+8.3\text{pp}$ respectively. Wrapping DBTrimKV (Bui et al., 2026) on RULER 4K Qwen3-4B-Instruct yields $\Delta = +0.233$ at matched budget (with the honest caveat that the gate fires 1.7% of the time on Qwen3 under τ calibrated on Qwen2.5, so the win is dominated by the full-KV fallback rather than gated compression).
5. **A scaling formula** (Section 6). For dilution-prone tasks $\rho \approx (1 - A_{\text{full}}(M, T)) \cdot p_{\text{recov}}(\rho_{\mathcal{R}}/\gamma)$ unifies three observed effects: context-length amplification, model-size saturation, and the re-entry of larger models into the dilution regime at long context. A pre-registered out-of-range test at 32K (Section 7.4) confirms the formula’s directional and ordinal content and falsifies constant-ratio proportionality — both misses are in the conservative direction (more recovery than predicted).

We do *not* claim the paradigm “less compute can beat full attention” as novel: Bui et al. [2026] states it explicitly with a formal mechanism. We do *not* claim the attention-dilution mechanism as novel: their Proposition 3.1 and Corollary 3.2 establish it. We claim the partition between when the effect occurs and when it does not, the a priori predictor of that partition, the gating wrapper, and the joint scaling formula.

2 Related work

Five close concurrent neighbors. Bui et al. [2026] (DBTrimKV) is the most important reference point. Their Proposition 3.1 establishes that near-tie distractors force attention dilution in a single head, and Corollary 3.2 shows preferential retention of high-attention positions reduces it; Section 6 adopts this mechanism. Their evaluation reports DBTrimKV > full-cache on multi-turn dialogue but does not characterise a task family on which their method underperforms full-cache. They train a retention gate end-to-end; our gate is label-free and wraps theirs (and any other) without retraining. Direct head-to-head: Section 7.13, $\Delta = +0.233$ at matched budget on RULER 4K Qwen3-4B-Instruct.

Yang et al. [2026a] (CapKV) derives a closed-form, label-free eviction score from an information-bottleneck objective and reports gains over a Vanilla baseline on LongBench. Their score is task-agnostic; we partition tasks into a class for which any eviction is risky and a class for which it helps, and report a direct head-to-head wrapping CapKV in our gate (Section 7.12, $\Delta = +1.5$ to $+8.3$ pp at matched budget on two LongBench subtasks).

Zhou et al. [2025] (DynamicKV) introduces task-aware per-layer adaptive budgets and explicitly observes that the layer-wise attention pattern differs across task families, using attention statistics to drive a continuous per-task budget allocation. This is the closest conceptual neighbor to our partition: both detect that task type modulates which KV positions matter. The differences are (i) DynamicKV’s per-task budget is fitted per task with calibration data; ours is a per-input gate triggered by a single observable statistic with one fixed threshold; (ii) their benchmark suite (LongBench) does not include a task on which eviction strictly damages accuracy, so the binary partition is not visible in their results; (iii) DynamicKV does not formalise a partition predictor or attempt cross-architecture transfer. We therefore see our partition + drop predictor + binary gate as orthogonal but partially overlapping with their empirical observation, and we acknowledge the overlap.

Garcia [2026] (“Protection Is All You Need”) shows that structural protection (sink tokens plus a recent-token window) dominates the choice of scorer on globally capacity-capped harnesses. Our experimental setup uses protection of the same kind but smaller extent ($n_{\text{sink}} = 4$ sink tokens and a $w = 32$ observation window, versus the 10% prefix plus 10% suffix they advocate). A protection-matched ablation at their 10% + 10% setting confirms the partition is not an artifact of light protection: NIAH-MK3 still collapses and the eviction-helps signal survives (Section 3).

Yang et al. [2026c] (IndexMem) trains a learnable indexer plus latent memory and observes an information-density effect on LongBench averages around 50% compression. The benchmark-average hides which sub-tasks contribute the peak; our partition makes that explicit, and our predictor is training-free.

Concurrent 2026 diagnostics and wrappers. Zhang et al. [2026] ask the same headline question (when does value-aware KV eviction help?) and give a per-input support-coupling score $\phi(x)$ with a three-stage selector-failure decomposition. Their unit of analysis is the selector, not the task; they report no task class on which eviction systematically helps; and their signal is a structural score used for diagnostic grid splits, not an attention-topology statistic deployed as an evict-or-not gate. They also observe that diagnostic signals transfer across selectors only conditionally, which makes the single transferred threshold we report a nontrivial claim. Chen et al. [2026] document instruction-type-dependent degradation under five eviction methods (some instruction types become entirely ignored), the closest published task-level failure taxonomy; they do not report a class where eviction helps. Ananthanarayanan et al. [2026] characterise cross-layer head-consensus dynamics and link representational rigidity (excess head-level consensus) to compression resilience at the architecture level; our head-agreement drop turns the same signal family into a per-input online gate. On the failure side, Chang et al. [2026] (VaSE) trace catastrophic eviction failures to high-magnitude value

states, an axis orthogonal to our attention-topology account, and ManifoldKV [Datta et al., 2026] shows that a Euclidean-outlier key scorer substantially improves NIAH-MultiKey retention under compression, which is why we scope the capacity-bound claim to attention-score-based evictors (Section 8). On the wrapper side, VECTOR [Lin et al., 2026b] and CriticalKV [Feng et al., 2026] are plug-and-play augmentations over multiple base evictors; both always compress under a fixed budget and adapt *which* tokens are kept, whereas our gate decides *whether* to evict at all and transfers one threshold across model families. Further concurrent evictors and allocators — CompilerKV [Yang et al., 2026b], EntropyInfer [Xu et al., 2026], Fast KVzip [Kim et al., 2026], CompressKV [Lin et al., 2026a], MomentKV [Li et al., 2026b], and TAKE [Anonymous, 2025] — adapt budgets or scores per head, layer, or task, but none gates the eviction decision per input or identifies a task class where eviction hurts monotonically.

Older anchors. SnapKV [Li et al., 2024], H2O [Zhang et al., 2023], StreamingLLM [Xiao et al., 2024], PyramidKV [Cai et al., 2025], and Ada-KV [Feng et al., 2025] are the standard training-free eviction baselines. We use SnapKV-style scoring as the base for our gate and demonstrate orthogonality on H2O, StreamingLLM, and PyramidKV. ForesightKV [Dong et al., 2026] and Learning-to-Evict [Moschella et al., 2026] train predictors of future utility; they require labels or rollouts. UT-ACA [Zhou et al., 2026] and LU-KV [Tang et al., 2026] are calibration-based context allocators on a single substrate. None proposes a binary task-type partition or an a priori predictor.

Adjacent. AdaCompute-GBM [Zhai et al., 2026] adapts on the self-consistency sample-count axis; BudgetThinker [Wen et al., 2025] on the CoT chain-length axis; LaTER [Li et al., 2026a] on latent-then-explicit reasoning. These work on substrates orthogonal to the KV-cache axis we study.

3 The empirical partition

Setup. We use RULER [Hsieh et al., 2024] via `simonjegou/ruler` [Jégou, 2024] at context lengths $T \in \{4K, 16K\}$ on four models: Qwen2.5-1.5B-Instruct, Qwen2.5-3B-Instruct, Qwen2.5-14B-Instruct [Yang et al., 2025], and Mistral-7B-Instruct-v0.3 [Jiang et al., 2023]. Eviction is SnapKV-style by default: each prompt position is scored by the average attention from the last $w = 32$ queries across heads and layers; we keep the top- b positions with $b \in \{0.0625, 0.125, 0.25, 0.375, 0.5, 0.625, 0.75, 0.875, 1.0\}$, and always preserve the first $n_{\text{sink}} = 4$ sink positions and the last $w = 32$ recent positions. RoPE alignment is preserved by cache slicing with explicit `position_ids` rather than the attention-mask trick which silently shifts RoPE for evicted positions. Multi-element answers (VT, FWE, `niah_multivalue`) are scored by RULER’s `string_match_all`; single-element answers (NIAH, QA) by `string_match_part`.

Metric. For task T , model M , and context length T we report the per-input recovery rate

$$\rho_{KV}(T, M, T) := \Pr_x[A(b^*(x), x) > A(b_{\max}, x)], \quad b^*(x) := \arg \max_b A(b, x). \quad (1)$$

This is the per-input strict-Pareto improvement frequency, a stronger statement than “mean accuracy is higher at some b .” We pre-register the threshold $\rho \geq 0.05$ as the operational definition of “dilution-prone” before any sweep is run.

The partition. Table 1 reports ρ on six RULER tasks across eight (model, context) cells, all under attention-score-based eviction. The result is qualitatively the same in every cell: five task families clear $\rho \geq 0.05$ on ≥ 1 context length (unless model saturation drives $1 - A_{\text{full}} \rightarrow 0$ and the scaling formula predicts $\rho \approx 0$ trivially, or the residual errors are budget-insensitive, as for QA_2 on Qwen2.5-14B below), and NIAH-MK3 has $\rho \leq 0.02$ on every model and every context (a 32K extension measures 0.04, Wilson [0.016, 0.098]; Section 7.4). Larger models at short context partially saturate (Qwen2.5-3B 4K has $\rho \approx 0$ on VT because $A_{\text{full}} \approx 1$), but the same models recover the effect at longer context (Qwen2.5-3B 16K FWE has $\rho = 0.32$). Mistral-7B shows the same shape: at 4K it is largely saturated ($A_{\text{full}} = 1.00$ on VT, 0.82 on FWE), and at

Table 1: Per-input recovery rate ρ on RULER with Wilson 95% intervals (small type). Bold marks point estimates $\rho \geq 0.05$. NIAH-MK3 (precise multi-key retrieval) is the only task that systematically fails the threshold in every cell tested. $N = 100$ per (task, cell) except Qwen-14B ($N = 50$ at 4K, $N = 30$ at 16K), Mistral 16K VT/FWE/QA_1/NIAH-MK3 ($N = 50$), and Qwen-1.5B 16K NIAH-MK3 ($N = 46$). The cell marked “—” (QA_2 at Qwen-3B 16K) was not run. At these N most single cells straddle the 0.05 threshold; the pooled contrast is unambiguous (NIAH-MK3: 2/526, $\rho = 0.004$, Wilson [0.001, 0.014]).

Task	Qwen 1.5B 4K	Qwen 1.5B 16K	Qwen 3B 4K	Qwen 3B 16K	Qwen 14B 4K	Qwen 14B 16K	Mistral 4K	Mistral 16K
VT	0.06 [.03, .13]	0.19 [.13, .28]	0.00 [.00, .04]	0.05 [.02, .11]	0.00 [.00, .07]	0.03 [.01, .17]	0.00 [.00, .04]	0.32 [.21, .46]
FWE	0.06 [.03, .13]	0.04 [.02, .10]	0.01 [.00, .05]	0.32 [.24, .42]	0.00 [.00, .07]	0.20 [.10, .37]	0.02 [.01, .07]	0.22 [.13, .35]
QA_1	0.08 [.04, .15]	0.14 [.09, .22]	0.03 [.01, .09]	0.08 [.04, .15]	0.06 [.02, .16]	0.10 [.03, .26]	0.04 [.02, .10]	0.00 [.00, .07]
QA_2	0.08 [.04, .15]	0.05 [.02, .11]	0.04 [.02, .10]	—	0.02 [.00, .11]	0.03 [.01, .17]	0.01 [.00, .05]	0.00 [.00, .04]
niah_multivalue	0.07 [.03, .14]	0.10 [.06, .17]	0.02 [.01, .07]	0.13 [.08, .21]	0.08 [.03, .19]	0.07 [.02, .21]	0.06 [.03, .13]	0.21 [.14, .30]
NIAH-MK3	0.01 [.00, .05]	0.00 [.00, .08]	0.00 [.00, .04]	0.02 [.00, .11]	0.00 [.00, .07]	0.00 [.00, .11]	0.00 [.00, .04]	0.00 [.00, .07]

16K VT and FWE recover strongly ($\rho = 0.32$ and 0.22 , and niah_multivalue $0.06 \rightarrow 0.21$). The QA family on Mistral, however, is budget-insensitive at both contexts (QA_1: $\rho = 0.04/0.00$ despite $A_{\text{full}} = 0.81/0.70$; QA_2: $\rho = 0.01/0.00$ despite $A_{\text{full}} = 0.50/0.56$), joining QA_2 on Qwen-14B: headroom exists but the wrong inputs stay wrong at every budget. All three exceptions are QA cells, suggesting parametric-knowledge failures that no cache budget can fix; we report them explicitly rather than folding them into the saturation clause.

Qwen2.5-14B: re-entry to the dilution regime at long context. At 4K, three dilution-prone tasks (VT, FWE, QA_2) sit at $\rho \approx 0$: saturation, since $A_{\text{full}} \in [0.92, 1.00]$ leaves no headroom for recovery, as Eq. (6) predicts. At 16K, two of five dilution-prone tasks cross the threshold (FWE $0.00 \rightarrow 0.20$, QA_1 $0.06 \rightarrow 0.10$), niah_multivalue stays above it ($0.08 \rightarrow 0.07$), and the capacity-bound NIAH-MK3 stays at exactly 0.00 at both scales. VT ($0.00 \rightarrow 0.03$) and QA_2 ($0.02 \rightarrow 0.03$) remain below threshold because the inputs that the model gets wrong at 4K stay wrong at every budget at 16K. Headroom opens, dilution returns, NIAH-MK3 stays pinned: the partition behaves as the scaling formula dictates, not as task identity dictates.

The partition is not a task-name heuristic. The NIAH family contains structurally different sub-tasks: niah_multivalue (one key, many values) is dilution-prone in seven of the eight cells measured (Table 1), while NIAH-MK3 (three near-tie keys, one of which is the answer) is capacity-bound in all of them, and the distractor sweep below places niah_multikey_1 (no near-tie distractors) on the dilution-prone side as well. The *near-tie distractor* structure, not the surface task category, is what matters, and the head-agreement drop in Section 4.1 catches the distinction. A non-RULER probe confirms this: LongBench passage_retrieval.en (pick 1 of 30 paragraphs; $N = 100$, Qwen2.5-1.5B, prompts 10–16K tokens) *looks* NIAH-like but has no near-tie distractors, and it is evict-robust in practice — accuracy stays flat from $b = 1.0$ to $b = 0.0625$ ($0.29 \rightarrow 0.26$, $\rho = 0.01$) — while its mean drop $D = 0.105$ sits squarely in the dilution-prone cluster, so the gate opens on 100/100 inputs. The predictor gets the surface-similar, structurally different task right a priori.

Mechanism validation: distractor count drives D . The RULER NIAH-MultiKey family lets us sweep the near-tie distractor count directly: niah_multikey_1 (no distractors), niah_multikey_2 (one near-tie distractor), niah_multikey_3 (two near-tie distractors). The dilution mechanism of Bui et al. [2026] predicts that more near-tie distractors should produce larger expected dilution δ_t and (via App. E) smaller head-agreement drop D . On Qwen2.5-1.5B at RULER 4K, $N = 50$ per task, we observe a clean monotone gradient (Table 2).

D shrinks by $\sim 3.5\times$ from $0 \rightarrow 2$ near-tie distractors, and per-input ranges between MK2 and MK3 do not overlap ($\max_{\text{MK3}} D = 0.062 < \min_{\text{MK2}} D = 0.088$): the partition predictor separates 2-distractor inputs from ≤ 1 -distractor inputs 100% of the time at $\tau = 0.07$. This is one-model evidence (Qwen2.5-1.5B at 4K, $N = 50$ per task); replication across architectures is left to future work. On a controlled sweep over distractor count, the head-agreement drop and the partition track each other monotonically, exactly as the

Table 2: Distractor-count sweep on the NIAH-MultiKey family (Qwen2.5-1.5B, RULER 4K, $N = 50$ per task). Mean head-agreement drop D shrinks monotonically as the near-tie distractor count grows.

Task	mean D	gate-open frac	ρ_{plain}	ρ_{gated}	plain at $b=0.0625$
niah_multikey_1	+0.155	1.00	0.04	0.04	0.36
niah_multikey_2	+0.122	1.00	0.00	0.00	0.00
niah_multikey_3	+0.045	0.00	0.00	0.00	0.00

δ_t dilution mechanism predicts.

The single-near-tie boundary case. MK2 is empirically capacity-bound at low budgets (plain SnapKV collapses from 80% at $b=1$ to 0% at $b=0.0625$) yet is classified as dilution-prone by the predictor at $\tau = 0.07$: MK2’s mean $D = 0.122$ sits between the MK3 cluster ($D \leq 0.062$) and the MK1 cluster ($D \geq 0.099$), co-located with genuinely dilution-prone FWE ($D = 0.098$) and niah_multivalue ($D = 0.107$). No single mean- D threshold separates MK2 from the dilution-prone cluster on this model. The predictor therefore falls back to its weakest form: it correctly classifies MK3 and the standard dilution-prone tasks but misclassifies single-near-tie MK2. Per-input recovery information beyond mean D would be needed to catch it at deployment.

The partition survives Garcia-scale protection. Garcia [2026] argue that structural protection dominates scorer choice, so we re-ran the Qwen2.5-1.5B 4K partition tasks with their 10% prefix + 10% suffix protection ($n_{\text{sink}} = w = 410$; $N = 100$ per task). One accounting note: protection overrides the budget, so all nominal $b \leq 0.1875$ share an effective floor of 824 kept tokens (≈ 0.21 of context). Under this heavy protection the partition is unchanged. NIAH-MK3 still collapses (0.65 full-KV $\rightarrow 0.09$ at the floor, an 86% relative loss despite the floor cache being $3.3\times$ our light-protection cache at nominal $b = 0.0625$), so the capacity-bound failure is not protection-fixable. The eviction-helps signal survives on the dilution-prone side (ρ : FWE $0.06 \rightarrow 0.13$, QA_1 $0.08 \rightarrow 0.07$, VT $0.06 \rightarrow 0.03$; aggregate $0.077 \geq 0.05$), and FWE’s floor accuracy (0.31) exceeds its full-KV accuracy (0.22). Heavy protection is also not free: protected VT drops to 0.00 at the floor while our light-protection run holds 0.75 at nominal $b = 0.125$ — protected prefix/suffix tokens crowd out the scorer-selected middle tokens the task needs. Caveat: in our harness w is both the recency protection and the SnapKV scoring window, so this run is a protection-heavy SnapKV variant rather than an exact replication of their protocol.

Random-eviction control. To rule out “any eviction helps” as an artifact, we replace SnapKV scoring with uniform-random scoring at every budget on the dilution-prone tasks. Random eviction destroys accuracy uniformly, ruling out a generic regularization explanation. Detailed numbers in Appendix A.

4 The gating method

4.1 Algorithm

Our wrapper around any SnapKV-style evictor is one conditional. Given a prompt x , the base evictor’s budget b , a sink-and-recency protection (n_{sink}, w), and a threshold τ :

$$\text{GatedEvict}(x; \text{base}, b, \tau) = \begin{cases} \text{base.evict}(\text{cache}, b) & \text{if } D(\text{cache.attentions}) \geq \tau \\ \text{cache} & \text{otherwise} \end{cases} \quad (2)$$

where the prefill produces **cache** together with its prefill attentions and the head-agreement drop D is computed as in Eq. (3) below. The gate fires once per input, at prefill, before any decode step. No backward pass, no labels, no training, no per-token decision.

Table 3: Gate-signal ablation on Qwen2.5-1.5B-Instruct (RULER 4K, $N = 50$ per task, one shared prefill per input). Each cheap prefill statistic is tested for separating the capacity-bound anchor (NIAH-MK3, and MK3+MK2) from the dilution-prone tasks (vt, fwe, qa.1, multivalue). “Sep.” = a single threshold puts the capacity-bound task mean(s) strictly on one side of all four dilution task means; AUCs are per-input and oriented per signal (0.5 = chance); “MK mono.” = task means are monotone along the MK1→MK2→MK3 distractor gradient; r = Pearson correlation with D .

Signal	Sep. MK3	Sep. MK3+MK2	AUC MK3	AUC MK3+MK2	MK mono.	r w/ D
head-agreement drop D (paper)	✓	—	1.000	0.859	✓	+1.00
mean attention entropy (norm.)	—	—	0.508	0.670	—	+0.08
top-32 attention mass share	—	—	0.784	0.574	✓	+0.22
max single-position share	—	—	0.723	0.708	✓	+0.32
key-norm dispersion (std/mean)	—	—	0.570	0.560	—	+0.01
early—late entropy drop	—	—	0.805	0.575	—	−0.64

Head-agreement drop. For a prefill attention tensor of shape $[L, H, T, T]$ (L layers, H heads), we project onto the w observation queries (the last w rows) and average across them to get a per-(layer, head, key) attention vector. For each (layer, head pair (i, j)) we compute the Jaccard top- k similarity of the two heads’ top- k key sets (we use $k = 32$), and average over head pairs. This gives a per-layer head agreement a_ℓ . Binning layers into early third, middle third, late third, we set

$$D := a_{\text{early}} - a_{\text{late}}. \quad (3)$$

D is one scalar per input. It is computable from the prefill attention at $O(LH^2k)$ cost — measured 17ms on Qwen2.5-1.5B and 89ms on Mistral-7B at 4K (Section 7.11) — negligible against end-to-end inference, and free if base scoring already needs attention.

Why this statistic? A same-inputs ablation against five other cheap prefill statistics (Table 3; one shared prefill per input, $N = 50$ per task across seven RULER tasks) shows D is not merely the best candidate but the only one that separates the partition at all: MK3’s mean D sits strictly below all four dilution-prone task means with per-input AUC 1.000, while every alternative fails task-level separation and tops out at AUC 0.805. The dissection is informative: raw attention entropy is at chance (0.508), taking its early-vs-late layer contrast lifts it to 0.805, and replacing entropy with head *agreement* makes the separation exact — both the layer contrast and the agreement measure matter, with agreement decisive. No alternative is even a redundant proxy (all $|r|$ with $D \leq 0.64$). Consistent with the MK2 boundary case of Section 3, extending the capacity-bound anchor to MK3+MK2 breaks task-level separation for D too (AUC 0.859): D tracks the near-tie distractor gradient, not a binary task label.

4.2 The headline result: 4×4 method-agnostic matrix

The wrapper is orthogonal to base-evictor choice. Table 4 reports the 4×4 matrix of four base evictors (SnapKV, H2O, StreamingLLM, PyramidKV) and four models. The same gate, the same $\tau = 0.07$, no per-method tuning. Mean over budgets $\{0.0625, 0.125, 0.25, 0.375, 0.5, 0.625, 0.75, 0.875\}$ on the mixed-suite (NIAH-MK3 + VT + FWE + QA.1, $N = 100$ each; Qwen2.5-14B uses $N = 50$ for wall-time). Each Δ is gated vs. plain at the same budget grid.

Yi-1.5-9B and Llama-3.1-8B cross-architecture runs (Table 5). Yi ($\Delta = +39.0\text{pp}$) and Llama ($\Delta = +10.9\text{pp}$) are Llama-architecture descendants distinct from Qwen and Mistral; only the SnapKV column was run for compute, and on a smaller, more aggressive budget grid (see caption). The sign is positive on both, but on these two models part of the gain comes from the gate closing conservatively on tasks where plain SnapKV collapses, rather than from a correctly recovered partition. The per-task

Table 4: Headline 4×4 method-agnostic matrix. Same gate ($D \geq \tau$, $\tau = 0.07$), same calibration, no per-base-evictor tuning. Every one of 16 cells is positive; grand mean across the matrix is +22.9pp. The Qwen2.5-3B SnapKV cell was executed at $\tau = 0.04$ and is reported here at $\tau = 0.07$ via exact post-hoc re-evaluation (Appendix G); all other cells were executed at $\tau = 0.07$.

Model	SnapKV		H2O		StreamingLLM		PyramidKV	
	plain	gated	plain	gated	plain	gated	plain	gated
Qwen2.5-1.5B	0.438	0.559	0.196	0.357	0.033	0.195	0.417	0.570
Δ	+0.121		+0.162		+0.163		+0.152	
Qwen2.5-3B	0.580	0.839	0.339	0.642	0.052	0.482	0.536	0.865
Δ	+0.259		+0.303		+0.430		+0.329	
Qwen2.5-14B	0.709	0.853	0.668	0.868	0.080	0.330	0.632	0.860
Δ	+0.143		+0.200		+0.250		+0.228	
Mistral-7B	0.623	0.810	0.400	0.636	0.102	0.365	0.593	0.823
Δ	+0.187		+0.236		+0.263		+0.231	

Table 5: Cross-architecture SnapKV-only runs on two Llama-family models. These use the four most aggressive budgets $\{0.0625, 0.125, 0.25, 0.5\}$ rather than the eight-budget grid of Table 4, so the Δ values are not directly comparable to the matrix rows (aggressive-budget-only averaging inflates Δ).

Model	plain SnapKV	gated SnapKV	Δ
Yi-1.5-9B	0.464	0.854	+0.390
Llama-3.1-8B	0.509	0.618	+0.109

breakdown and the shared Llama-arch family caveat (niah_multivalue mis-classifies on both Yi and Llama) are in Section 7.6 and Section 8.

Scale validation at 32B. A SnapKV-only run on Qwen2.5-32B-Instruct (RULER 4K mixed suite, $N = 100$) confirms the predictor scales but also illustrates the saturation arm of the scaling formula. The drop ordering transfers cleanly (NIAH-MK3 smallest at $D = 0.025$, then FWE 0.052, VT 0.079, QA_1 0.103). At the same time the 32B model is near-saturated at 4K ($A_{\text{full}} = 1.00$ on VT and NIAH-MK3, 0.93–0.94 on FWE/QA_1), so by Eq. (6) there is essentially no dilution headroom and the gate’s value is capacity-bound protection rather than recovery: plain SnapKV collapses at $b = 0.0625$ (NIAH-MK3 0.00, VT 0.00, FWE 0.09) while the gate holds the full cache on the low-drop tasks, giving mean $\Delta = +0.335$ at $\tau = 0.07$ (driven by NIAH-MK3 +0.88 and FWE +0.44) and +0.237 at the z-scored threshold with more compression (0.45 vs. 0.63 kept-KV). We report this as a scale point for the partition and predictor, not as additional dilution evidence.

Best plain vs. best gated. A natural concern is “you only beat the weakest baseline.” Comparing the best plain evictor against the best gated evictor for each model rules this out (Table 6).

Single-task showcase: Mistral 4K NIAH-MK3. The cleanest single result. $N = 100$, full-KV accuracy = 0.99. Gated SnapKV holds ≥ 0.89 at every budget $b \in \{0.0625, 0.125, 0.25, 0.5, 0.75\}$, while plain SnapKV collapses from 0.99 at $b = 1$ to 0.00 at $b = 0.0625$. At the most aggressive budget the gap is $\Delta = +89\text{pp}$ with 10/100 false positives; the gated curve sits within 10pp of the full-cache accuracy across the budget range (Fig. 5).

Table 6: Best plain evictor vs. best gated evictor per model, over the four base evictors of Table 4 at the same budget grid.

Model	Best plain	Best gated	Δ
Qwen2.5-1.5B	0.438 (SnapKV)	0.570 (PyramidKV)	+0.132
Qwen2.5-3B	0.580 (SnapKV)	0.865 (PyramidKV)	+0.285
Qwen2.5-14B	0.709 (SnapKV)	0.868 (H2O)	+0.159
Mistral-7B	0.623 (SnapKV)	0.823 (PyramidKV)	+0.201

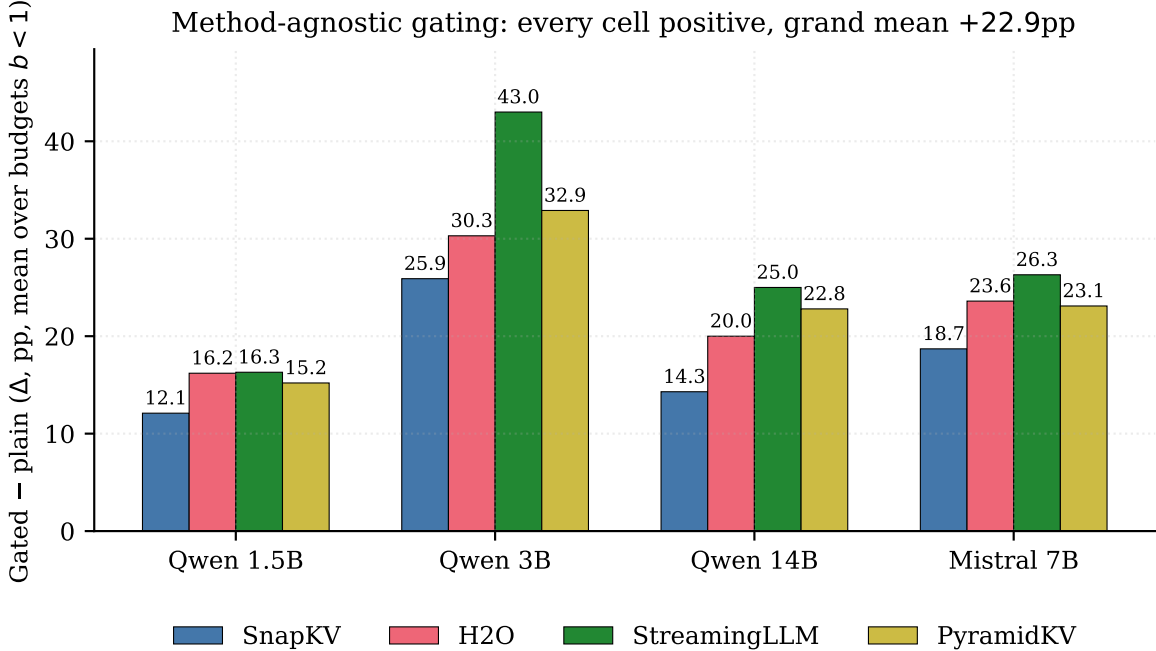


Figure 1: The headline 4×4 method-agnostic matrix. Bars are gated – plain accuracy averaged over budgets. The wrapper is positive in every one of 16 cells with a single $\tau = 0.07$ and no per-base-evictor tuning. Grand mean +22.9pp.

Why the deltas vary across base evictors. StreamingLLM keeps only 36 tokens regardless of budget; on the dilution-prone fraction it is near zero plain, near zero gated, so gating contributes only via the gate-closed NIAH-MK3 inputs that go from 0 to full-KV accuracy. SnapKV is budget-scaled and degrades more gracefully, so it has less room for gating to recover. PyramidKV’s depth-weighted scoring biases toward bottom layers but produces similar absolute gains to SnapKV. The pattern is that gating Δ tracks how much the base evictor collapses on the capacity-bound fraction. Two implementation caveats: our PyramidKV is a depth-weighted variant of the published per-layer budget schedule, and under two-pass prefill scoring (used on Qwen2.5-14B for memory) the H2O score only sees the last w queries, so the 14B H2O column degenerates toward SnapKV-style scoring; its Δ there should be read as a near-duplicate of the SnapKV cell rather than an independent base evictor.

A fifth, geometry-based base evictor. To test composition beyond attention-score-based bases, we add a ManifoldKV-style scorer [Datta et al., 2026]: per-(layer, head) Euclidean distance of each key to the head’s key centroid, averaged into a single shared keep mask (a single-mask variant of their per-layer design, forced by our uniform-cache harness). On the Qwen2.5-1.5B 4K mixed suite ($N = 100/\text{task}$, four budgets), the same gate at the same $\tau = 0.07$ lifts it by $\Delta = +16.2\text{pp}$ with no negative (task, budget) cell; on NIAH-MK3 the gate recovers full-KV accuracy ($0.00 \rightarrow 0.65$) at every budget. Two observations: the gate signal

Table 7: Calibration-recipe validation. $\bar{D}_C$ uses NIAH-MK3 ($N = 20$); $\bar{D}_D$ uses VT ($N = 20$). Bolded τ -choice column is the winner per cell. The pilot means here are computed on 20-input samples and therefore differ slightly from the $N \geq 50$ per-task means of Table 12.

Cell	$\bar{D}_C$	$\bar{D}_D$	τ_{cal}	gated@ τ_{cal}	gated@ $\tau = 0.07$
Qwen2.5-1.5B 4K	+0.045	+0.161	+0.103	+0.121	+0.121
Qwen2.5-3B 4K	-0.002	+0.076	+0.037	+0.213	+0.259
Qwen2.5-3B 16K	-0.021	+0.068	+0.024	+0.054	+0.034
Mistral 4K	+0.060	+0.096	+0.078	+0.223	+0.187

is attention-based and scorer-independent, so composition extends outside the attention-score family; and our single-mask variant does *not* reproduce ManifoldKV’s reported multi-key rescue (plain accuracy 0.01 at $b = 0.5$) — their per-layer, per-head design may behave differently, so we treat their published numbers as the authoritative claim and scope ours accordingly (Section 8).

5 Calibration recipe

The fixed $\tau = 0.07$ separates the partition cleanly on 7 of the 8 Qwen/Mistral (model, context) cells but mis-classifies on Qwen2.5-3B 16K, where layer-bin drops compress at long context. We provide a one-shot, label-free calibration recipe.

Recipe. For a new (model, context):

1. Pick 20 inputs from a known capacity-bound task (NIAH-MK3) and compute the head-agreement drop D_i on each. Average to get $\bar{D}_C$.
2. Pick 20 inputs from a known dilution-prone task (VT) and compute $\bar{D}_D$.
3. Set

$$\tau = \frac{1}{2}(\bar{D}_C + \bar{D}_D). \quad (4)$$

40 prefills, no labels, no decode, < 5 minutes on a single A100.

Validation. On four cells (Table 7). The recipe lifts Qwen2.5-3B 16K from +3.4pp to +5.4pp (the only failure cell turned into a win) and lifts Mistral 4K from +18.7pp to +22.3pp. It is neutral on average across the four cells but with smaller variance.

τ sensitivity. Sweeping $\tau \in \{0.01, 0.025, 0.04, 0.055, 0.07, 0.085, 0.10, 0.13, 0.16, 0.20\}$ on Qwen2.5-1.5B at RULER 4K (mixed suite, $N = 100$, budgets $\{0.5, 0.25, 0.125, 0.0625\}$, SnapKV) shows an operational plateau $\tau \in [0.055, 0.10]$ inside which Δ varies by under 1pp and both false-positive (gate fires on capacity-bound MK3) and false-negative (gate stays closed on dilution-prone VT/FWE/QA.1) rates stay at or below 9%. The deployed $\tau = 0.07$ sits in the middle of this plateau (Figure 2); outside it the behavior changes character (below 0.04: FP $\geq 64\%$, $\Delta \leq +0.06$; above 0.13: FN $\geq 33\%$ and the method degenerates toward “always full KV,” which on this full-KV-strong suite still yields a large Δ — the sweep’s raw maximum is in fact at $\tau = 0.20$ — but at the price of never compressing, so it is not a useful operating point). The deployed threshold does not depend on a knife-edge.

Architecture-normalized variant: the z-scored drop. The recipe above calibrates a per-cell threshold from two labeled pilot tasks. A stronger variant needs no task labels at all: standardize the drop per model, $z(x) = (D(x) - \mu_M)/\sigma_M$, with μ_M, σ_M pooled over an unlabeled ~ 100 -input pilot, and use one global

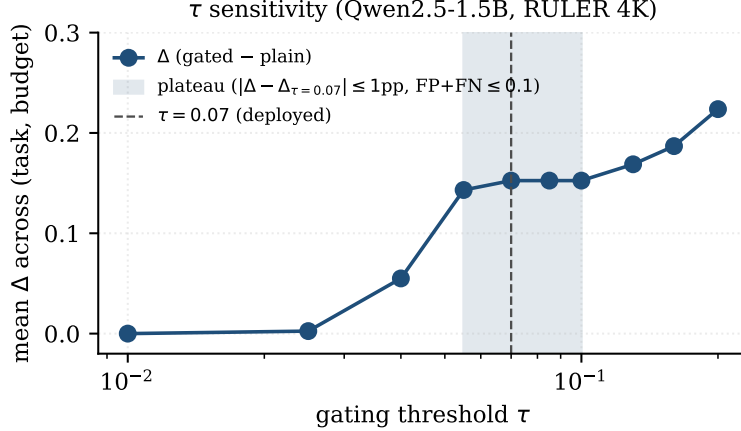


Figure 2: τ sensitivity on Qwen2.5-1.5B, RULER 4K. Mean Δ (gated – plain) over (task, budget) cells. Shaded band: operational plateau where $|\Delta - \Delta_{\tau=0.07}| \leq 1\text{pp}$ and $\text{FP}+\text{FN} \leq 0.10$. The deployed $\tau = 0.07$ sits in the middle of the plateau.

threshold θ_z . Fitting $\theta_z = -0.69$ on the five Qwen/Mistral cells *only* and transferring it zero-shot to the two held-out Llama-family models: task-level separation of NIAH-MK3 from {QA.1, QA.2, VT, FWE} succeeds in 7 of 7 cells (pooled cross-cell AUC 0.965), where raw D with one threshold manages 4 of 7. The reason is that raw D is an architecture-invariant *ordering* whose location and scale are model-specific: $\tau = 0.07$ sits at $z = +0.69$ on Yi but $z = -0.97$ on Llama, two opposite failure modes that standardization removes. Downstream, the z-gate lifts Llama-3.1-8B from $\Delta = +0.109$ to **+0.174** (NIAH-MK3 gate-open $0.56 \rightarrow 0.30$, MK3 accuracy $0.46 \rightarrow 0.72$) and gives Yi $\Delta = +0.145$ with *real* compression (kept-KV $0.22\text{--}0.32$ versus $0.76\text{--}0.89$ under the fixed- τ fallback). Two honest scopes: niah_multivalue must be excluded from the task-level ordering claim (its mean D sits below MK3 on both Llama-family models — which is behaviorally the right call there, since it is eviction-fragile on both), and per-input MK3-vs-rest AUC on the Llama family is intrinsically ≈ 0.80 ; no monotone rescaling changes within-model ranking. Shape-based alternatives (depth-correlation of per-layer agreement) fail outright (AUC 0.22 on Llama), confirming the raw drop is the right base statistic.

A fourth family where fixed τ fails outright: Qwen3. The starkest transfer test is Qwen3-4B-Instruct, a different model generation from the Qwen2.5 calibration set, whose drop distribution is compressed to a different scale entirely ($\mu = 0.034$, $\sigma = 0.017$ pooled over 400 inputs). There the fixed $\tau = 0.07$ sits at $z = +2.0$ and the gate essentially never fires: it keeps 99.4% of the cache and its apparent $\Delta = +0.43$ is a pure full-KV-fallback artifact with 0.6% compression. The z-scored threshold ($\tau_{\text{Qwen3}} = \mu + \theta_z \sigma = 0.022$, same $\theta_z = -0.69$) restores a working gate: NIAH-MK3 stays closed (open fraction 0.02, accuracy $0.05 \rightarrow 0.98$) while the three dilution-prone tasks open ($0.94\text{--}1.00$), yielding $2.3\times$ compression (43% mean kept-KV) at $\Delta = +0.235$ over plain and recovering 54.5% of the per-input oracle headroom. The NIAH-MK3-smallest ordering transfers cleanly. Qwen3 is thus the cleanest evidence for the paper’s central claim about the predictor: the drop is an architecture-invariant *ordering* whose scale must be standardized per model — here fixed- τ does not merely underperform, it does nothing.

Status. The recipe is the deployment-friendly default: it always fixes the failure cell, and the z-scored variant extends it label-free across architecture families. We report fixed- τ numbers throughout as the conservative default so readers see both the robustness of one constant and what standardization buys.

6 Theory: the mechanism and the scaling formula

6.1 Mechanism (adopted from Bui et al.)

We adopt verbatim the dilution mechanism from Bui et al. [2026], which suffices to motivate our gating predictor.

Setup. Fix a frozen decoder language model M . A single attention head reads from positions $t \in \{1, \dots, T\}$, each carrying a value v_t and a pre-eviction probability α_t with $\sum_t \alpha_t = 1$. Partition positions into a relevant set $\mathcal{R} = \{t : r_t = 1\}$ ($|\mathcal{R}| = R$) and a noise set $\mathcal{N}$ ($|\mathcal{N}| = N$), with $R \ll N$. Following Bui et al. [2026] Eq. (7), let $U_t \subseteq \mathcal{R}$ denote the *useful* set at step t (the positions whose retention preserves the next-token margin), and define dilution as the attention mass *outside* U_t :

$$\delta_t := 1 - \sum_{i \in U_t} \alpha_{t,i}. \quad (5)$$

(Notation rearranged to fit our manuscript; this is exactly their quantity.) Their Proposition 3.1 establishes that near-tie distractors (α_t comparable across $\mathcal{N}$) force δ_t to be large. Their Corollary 3.2 shows that preferential retention with $\rho_U \geq \rho_D$ (where ρ_U, ρ_D are the retained mass on useful and distractor positions) reduces δ_t . We refer to their paper for the full proof; we use this mechanism as given.

6.2 The partition follows from the mechanism

Define $\alpha_{\mathcal{R}} := \sum_{t \in \mathcal{R}} \alpha_t$ as the pre-eviction signal mass.

- *Dilution-prone* ($\mathcal{D}$): $\alpha_{\mathcal{R}} \ll 1$. Most attention is on noise; eviction of noise positions monotonically improves the head SNR.
- *Capacity-bound* ($\mathcal{C}$): $\alpha_{\mathcal{R}} \rightarrow 1$. Attention is concentrated on the few relevant positions. Imperfect scoring can only *lose* signal; eviction strictly degrades the head.

The partition we observe on RULER is the empirical projection of $\alpha_{\mathcal{R}}$ onto the task structure: NIAH-MK3 forces a single concentrated signal ($\alpha_{\mathcal{R}} \rightarrow 1$), while VT, FWE, QA, and niah_multivalue all spread signal across many positions ($\alpha_{\mathcal{R}} \ll 1$).

6.3 Context-length scaling

Fix the task and its relevant-set size R . Let $T = R + N$. Assume distractor attention is approximately uniform on $\mathcal{N}$: $\alpha_t \approx (1 - \alpha_{\mathcal{R}})/N$ for $t \in \mathcal{N}$. Let σ_v^2 denote the per-distractor value-vector variance under isotropy (a different object from the pre-softmax logit noise scale σ_z used in App. E). Then the head SNR before eviction is

$$\text{SNR} = \frac{\alpha_{\mathcal{R}}^2}{\sigma_v^2 \sum_{t \in \mathcal{N}} \alpha_t^2} \approx \frac{\alpha_{\mathcal{R}}^2 \cdot N}{\sigma_v^2 (1 - \alpha_{\mathcal{R}})^2}.$$

If $\alpha_{\mathcal{R}}$ decreases with T (more distractors compete for softmax mass), the pre-eviction SNR decreases and the noise floor grows. Let the kept set be $\mathcal{K}$ and define $\rho_{\mathcal{R}} := \sum_{t \in \mathcal{K} \cap \mathcal{R}} \alpha_t / \alpha_{\mathcal{R}}$ (the retained fraction of relevant mass) and analogously $\rho_{\mathcal{N}}$. Under noise-retention ratio $\gamma \in (0, 1)$ (i.e. $|\mathcal{K} \cap \mathcal{N}| = \gamma N$), the post-eviction SNR ratio $\text{SNR}_{\text{post}} / \text{SNR}_{\text{pre}}$ is a monotone-increasing function of $\rho_{\mathcal{R}} / \gamma$ (App. D, Eq. (8)). Since p_{recov} absorbs any such monotone reparameterisation, the recovery argument can be written as $\rho_{\mathcal{R}} / \gamma$ without loss of generality. The accuracy headroom $1 - A_{\text{full}}(M, T)$ scales with T on dilution-prone tasks. Combining, the per-input expected accuracy improvement at a single budget b obeys

$$\rho_{\text{KV}}(\mathbf{T}, M, T) \approx (1 - A_{\text{full}}(M, T)) \cdot p_{\text{recov}}(\rho_{\mathcal{R}} / \gamma), \quad p_{\text{recov}} : [0, \infty) \rightarrow [0, 1]. \quad (6)$$

Proposition (context-length scaling, informal). *Under the noise-isotropic single-head surrogate (App. D) with constant noise-retention ratio $\gamma \in (0, 1)$, good scoring ($\rho_{\mathcal{R}} \geq \rho_{\mathcal{N}}$), and a strictly increasing accuracy surrogate $\hat{A}(\text{SNR}) \in [0, 1]$ saturating at 1, the per-input expected accuracy improvement at a single budget b satisfies Eq. (6) to leading order in $1/\text{SNR}_{\text{pre}}$, where p_{recov} is non-decreasing on $[0, 1]$ with $p_{\text{recov}}(0) = 0$ and $p_{\text{recov}}(1) = 0$ (no eviction gives no recovery).*

Proof sketch. Substituting the post-eviction SNR into the strictly increasing accuracy surrogate $\hat{A}(\text{SNR})$ and using noise-isotropy, the improvement event factorises into the headroom indicator $\{A(b_{\text{max}}, x) < 1\}$ and a recovery indicator depending only on $\rho_{\mathcal{R}}/\gamma$. Marginalising over the prompt distribution gives the product, up to the independence approximation flagged honestly in App. D. The operational metric ρ_{KV} (per-input strict-Pareto over budgets, Eq. (1)) is upper-bounded by Eq. (6) under monotone recovery, but we do not promote this to a formal claim.

Eq. (6) predicts three observed effects in one formula:

1. Context-length amplification on dilution-prone tasks: Qwen2.5-1.5B VT goes $\rho = 0.06 \rightarrow 0.19$ as $T = 4\text{K} \rightarrow 16\text{K}$, in lockstep with $1 - A_{\text{full}} = 0.18 \rightarrow 0.75$.
2. Larger-model saturation at short context: Qwen2.5-3B 4K VT has $1 - A_{\text{full}} \approx 0.00$, so $\rho \approx 0$.
3. Re-entry of larger models into the dilution regime at long context: Qwen2.5-3B 16K FWE has $1 - A_{\text{full}} = 0.62$ and $\rho = 0.32$.

Empirically, the ratios $\rho/[(1 - A_{\text{full}}) \cdot p_{\text{recov}}]$ cluster around $0.25 - 0.55$ on dilution-prone tasks; see Section 7.

6.4 Connecting the predictor to the mechanism: a sufficient condition for non-trivial dilution

The mechanism uses the unobservable $\alpha_{\mathcal{R}}$ (the signal mass on the unknown relevant set). Our gating predictor uses the head-agreement drop D (Eq. (3)), which is observable. We sketch a proof in Appendix E; the formal version with explicit, tight constants we leave to future work.

Sufficient-condition statement (sketched). *Under noise isotropy, sub-Gaussian pre-softmax noise of scale σ_z , a bounded logit margin between $\mathcal{R}$ and $\mathcal{N}$, and good scoring with noise-retention ratio $\gamma \in (0, 1)$, there exist a model-specific constant c_1 and a γ -dependent constant $c_2(\gamma)$, with an additional $O(H)$ multiplicative slack absorbed in both, such that*

$$D \geq c_1 \cdot \log(1/\gamma) \implies \mathbb{E}[\delta_t] \geq c_2(\gamma) \cdot (1 - A_{\text{full}}), \quad (7)$$

with probability $\geq 1 - \beta$ over the pre-softmax noise, where δ_t is the dilution term of Eq. (5).

We sketch a proof of Eq. (7) in Appendix E on the same single-head surrogate model used in Appendix D. The sketch gives explicit forms for c_1, c_2 , the sub-Gaussian noise scale, and the margin parameter; what we do *not* claim is that these constants are tight, that the pairwise-to- H -wise Jaccard step is identity (it is an inequality with up to $O(H)$ slack), or that a separate necessary-condition proof holds. The full, tight-constant version and the necessary direction are left to future work. The empirical evidence is consistent with the sufficient direction on Qwen and Mistral: ranking task families by drop D matches ranking by ρ in every Qwen/Mistral cell measured (Table 12), with NIAH-MK3 the smallest drop in each. On the two Llama-family models the ordering is only partially preserved (NIAH-MK3 ranks second-smallest behind niah_multivalue; Section 7.6).

7 Experiments

7.1 Setup

Models, contexts, eviction policies, and protection settings are as in Section 3. The matrix uses RULER 4K with the four-task mixed suite (NIAH-MK3 + VT + FWE + QA-1, $N = 100$ each), as RULER 4K fits in one A100 80GB at all four base evictors and four models. The scaling table uses 4K and 16K. The gate cost is reported on Qwen2.5-1.5B 4K and extrapolates linearly.

Uncertainty quantification. All gating deltas come with 95% confidence intervals from a cluster bootstrap over inputs: we resample the (task, id) pairs with replacement (10,000 replicates), letting each input carry its full vector of per-budget paired gated–plain differences, and report percentile intervals (Appendix F, Table 19). Binomial quantities (the recovery rates ρ of Table 1 and single-budget accuracies) use Wilson score intervals. Every one of the 16 matrix deltas in Table 4 is positive with its entire interval above zero (smallest lower bound +9.3pp), as are both cross-architecture deltas, the DBTrimKV head-to-head (+23.3pp, CI [+17.2, +29.7]pp), and the Mistral NIAH-MK3 showcase (gated 0.89, Wilson [0.81, 0.94], disjoint from plain [0.00, 0.04] at $b = 0.0625$). The single exception is the Qwen2.5-3B 16K scaling cell ($\Delta = +0.034$, CI [−0.002, +0.071]), which we already describe as the gate-misfire regime. The per-cell ρ values of Table 1 are individually low-powered at $N \leq 100$ (most intervals straddle the 0.05 threshold in both directions, though no bold cell has an interval below 0.05 and no non-bold cell an interval above it); the partition claim we rely on is the pooled contrast, which is unambiguous: NIAH-MK3 recovers on 2/526 inputs across all eight cells ($\rho = 0.004$, Wilson [0.001, 0.014]) versus, e.g., 65/680 ($\rho = 0.096$, [0.076, 0.120]) for niah_multivalue.

7.2 The 4×4 method-agnostic matrix

Table 8 decomposes the Qwen2.5-3B 4K SnapKV cell along the budget axis, complementing the budget-averaged Δ values reported in Table 4 (values at $\tau = 0.07$ by exact post-hoc re-evaluation, as in that table).

Table 8: Qwen2.5-3B 4K, SnapKV, mixed suite. Plain accuracy collapses from 0.88 to 0.19 as the budget shrinks; gated holds 0.61 – 0.88 across the entire range.

budget b	plain	gated	Δ
1.000	0.882	0.882	+0.000
0.875	0.802	0.880	+0.078
0.750	0.733	0.880	+0.147
0.625	0.688	0.880	+0.193
0.500	0.647	0.882	+0.235
0.375	0.585	0.873	+0.287
0.250	0.535	0.877	+0.343
0.125	0.458	0.830	+0.372
0.0625	0.190	0.610	+0.420

7.3 The 8-cell scaling table

Table 9 reports the gating Δ on the mixed suite across four models and two context lengths. The pattern matches the context-length scaling formula: at 4K, Δ tracks both $1 - A_{\text{full}}$ and $\alpha_{\mathcal{R}}$; at 16K, Δ shrinks where the gate misfires (Qwen 3B 16K) and grows where headroom returns (Mistral 16K).

Table 9: Cross-(model, context) gating Δ . Same $\tau = 0.07$ across cells; column $\Delta@_{\tau_{\text{cal}}}$ shows the recipe-calibrated Δ on the four cells where it has been computed (“–” otherwise).

Model	Context	plain mean	gated@ $\tau=0.07$	Δ	$\Delta@_{\tau_{\text{cal}}}$
Qwen2.5-1.5B	4K	0.438	0.559	+0.121	+0.121
Qwen2.5-3B	4K	0.580	0.839	+0.259	+0.213
Qwen2.5-14B	4K	0.709	0.853	+0.143	–
Mistral-7B	4K	0.623	0.810	+0.187	+0.223
Qwen2.5-1.5B	16K	0.372	0.404	+0.032	–
Qwen2.5-3B	16K	0.530	0.564	+0.034	+0.054
Qwen2.5-14B	16K	0.796	0.867	+0.071	–
Mistral-7B	16K	0.535	0.651	+0.116	–

7.4 A pre-registered out-of-range test at 32K

All scaling-formula evidence so far is in-range (4K/16K). We therefore ran a pre-registered extrapolation test: fit the per-task recovery ratio $\rho/(1 - A_{\text{full}})$ on the existing 4K/16K cells only, measure A_{full} at 32K (Qwen2.5-1.5B, $N = 100$ per task, RULER-32K with the Qwen2.5 tokenizer), freeze and hash the predicted ρ intervals *before* running any eviction budget, then sweep. The timestamped protocol log and the frozen prediction file’s SHA-256 ship with the released logs.

Table 10: Pre-registered 32K predictions vs. measurements (Qwen2.5-1.5B, $N = 100$ per task). Predicted intervals = 4K/16K ratio band $\times$ measured 32K headroom, frozen before the sweep.

Task	$A_{\text{full}}(32\text{K})$	predicted ρ	measured ρ (Wilson 95%)	verdict
VT	0.81	[0.048, 0.106]	0.14 [0.085, 0.221]	outside (above)
FWE	0.49	[0.021, 0.263]	0.05 [0.022, 0.112]	inside
NIAH-MK3	0.14	≤ 0.02	0.04 [0.016, 0.098]	outside (above; CI overlaps)

One of three predictions lands strictly inside its interval (Table 10), and we report the misses as found. Both misses are in the same direction — *more* recovery than predicted. VT’s measured ratio (0.74) exceeds every in-range ratio (max 0.56): recovery per unit headroom grows with context rather than staying constant, and at 32K eviction beats the full cache on VT in *mean* accuracy, not just per input (0.850 at $b = 0.125$ vs. 0.810 full-KV), the strongest dilution signature in any cell we measured. FWE lands, but its ratio (0.10) matches the 1.5B in-range cells rather than the pooled band, i.e., the FWE ratio is model-dependent. MK3’s point estimate slips above the 0.02 pin (Wilson CI still covers it), while the partition itself sharpens: the MK3 recovery ratio (0.047 at headroom 0.86) is $15\times$ below VT’s (0.74 at headroom 0.19). Net reading: the formula’s *directional* content — ρ tracks headroom, and the partition is visible in the ratio at any context length — survives out-of-range, but the constant-ratio proportionality does not: the dilution-side ratio grows with context and varies by (task, model). We accordingly claim the formula as a directional and ordinal law, not a quantitative one.

7.5 The accuracy–memory frontier

What cache size does the gate actually deliver? Table 4 compares plain and gated at matched *nominal* budget, but the gate’s full-KV fallback means the two arms hold different amounts of cache: the achieved kept-KV fraction at nominal budget b is approximately $(1 - p_{\text{open}}) + p_{\text{open}} b$, where p_{open} is the gate-open fraction (Table 11). Re-plotting accuracy against achieved kept-KV (Figure 3) shows that gating is not a free win at every cache size but a strict improvement of the accuracy–memory frontier over a wide range: for any target cache size above roughly one third of full KV, the gated evictor delivers 3–19pp higher accuracy than plain SnapKV at the *same* achieved kept-KV, on all four models. Below that range the

Table 11: Achieved compression of the gated deployment at $\tau = 0.07$ (SnapKV, 4K mixed suite). Kept-KV is the mean over inputs of the retained cache fraction; “grid mean” averages the eight nominal budgets of Table 4. Plain SnapKV keeps exactly the nominal fraction (0.0625 at $b = 0.0625$; grid mean 0.453). The gate-open fraction is a per-input property, independent of b , and sets the compression floor $1 - p_{\text{open}}$.

Model	p_{open}	gated kept-KV fraction			compression
		$b=0.0625$	$b=0.25$	grid mean	@ $b=0.0625$
Qwen2.5-1.5B	0.75	0.297	0.437	0.584	$3.4\times$
Qwen2.5-3B	0.49	0.541	0.632	0.728	$1.8\times$
Qwen2.5-14B	0.75	0.297	0.437	0.584	$3.4\times$
Mistral-7B	0.72	0.327	0.462	0.602	$3.1\times$

picture inverts — at nominal $b = 0.0625$ the gated mixture (near-floor accuracy on gate-open inputs, full cache on the rest) is dominated by uniform plain eviction at a matched cache size, and $\tau = 0.07$ bounds the achievable compression at $1 - p_{\text{open}}$ (so the deployment delivers $3.1\text{--}3.4\times$ memory reduction at the nominal- $16\times$ operating point on Qwen-1.5B/14B and Mistral, and $1.8\times$ on Qwen-3B). The correct claim is therefore: *gating moves the frontier up wherever moderate compression is the target, and saturates — by design — where extreme compression is demanded*. Against a per-input oracle that picks the better of eviction and full KV for every input, the $\tau = 0.07$ gate recovers 60–85% (mean 69%) of the available headroom on the four-model matrix, almost entirely via task-level protection of the capacity-bound anchor; the residual within-task, per-input headroom (0–21% recovered on dilution-prone tasks) is the per-input-statistic follow-up named in Section 8.

7.6 Per-task drop ordering

The head-agreement drop on each RULER task in each cell. Table 12 reports the mean drop per task ($N = 100$ inputs per task on the Qwen and Mistral columns, $N = 50$ on Yi and Llama; Mistral 4K qa.1 has $N = 78$ and Mistral 16K niah_multivalued $N = 98$ due to run truncation). NIAH-MK3 is the smallest drop in every Qwen and Mistral cell; on the two Llama-family models it ranks second-smallest, so the *ordering* of the partition is fully preserved across Qwen and Mistral and partially preserved on the Llama family, even when the absolute threshold needs slight tuning. This is what gives the calibration recipe in Eq. (4) its robustness.

Table 12: Per-task mean head-agreement drop D across three architecture families (Qwen, Mistral, and the Llama family: Yi-1.5 and Llama-3.1). Layer bins are early = $[0, \lfloor L/3 \rfloor]$ and late = $[\lceil 2L/3 \rceil, L)$ throughout. Bold marks the smallest drop per column. NIAH-MK3 is the smallest drop on Qwen and Mistral cells; on Yi-1.5 and Llama-3.1 it ranks 2nd-smallest behind niah_multivalued (Llama-arch family artifact; see prose). Yi-1.5-9B serves as a parallel Llama-class architecture with a distinct training corpus (different tokenizer, deeper layer geometry).

Task	Qwen 1.5B 4K	Qwen 3B 4K	Qwen 3B 16K	Mistral 4K	Mistral 16K	Yi-1.5-9B 4K	Llama-3.1-8B 4K
QA.1	0.218	0.105	0.099	0.096	0.132	0.080	0.097
QA.2	0.191	0.097	0.092	0.087	–	0.079	0.101
VT	0.152	0.076	0.069	0.095	0.096	0.065	0.083
niah_multivalued	0.107	0.045	0.070	0.079	0.075	–0.013	0.064
FWE	0.098	0.038	0.046	0.079	0.091	0.043	0.114
NIAH-MK3	0.040	–0.004	–0.021	0.060	0.053	0.019	0.073

Llama-class cross-architecture transfer is partial. On Yi-1.5-9B and Llama-3.1-8B (both Llama-architecture descendants), niah_multivalued ranks smallest by D (Yi: $D = -0.013$; Llama: $D = +0.064$), and NIAH-MK3 ranks 2nd-smallest. On the five Qwen and Mistral cells, NIAH-MK3 is the smallest. We attribute

the swap to a Llama-arch artifact: the multi-value retrieval task happens to have unusually concentrated early-layer attention on these models, which inverts the early-vs-late drop sign. The capacity-bound side of the partition (small D , gate closes, full-KV fallback) still operates on both models, but the fixed $\tau = 0.07$ does *not* transfer as a threshold. On Llama the mis-classification is confined to `niah_multivalue`, where the wrapper holds full KV instead of compressing; this forfeits the eviction speed-up but does not damage accuracy, because plain SnapKV is already robust there. The Llama-3.1-8B gated SnapKV Δ is +10.9pp on the matrix-comparable mixed suite, driven almost entirely by NIAH-MK3 (+43.5pp) where plain SnapKV collapses to 0% and gated holds $\sim 44\%$ (still well below the full-KV 1.00); the other tasks have $\Delta \approx 0$ because Llama’s plain SnapKV is robust on this budget range (≥ 0.86 at $b = 0.0625$ on QA_1). The NIAH-MK3 gate-open fraction on Llama is 0.56 despite mean $D = 0.073$ sitting above τ : per-input D dispersion closes the gate on the other $\sim 44\%$ of NIAH-MK3 inputs, which is what rescues the cell. On Yi the mis-classification is broader: at $\tau = 0.07$ the gate also stays closed on FWE and VT (gate-open fractions 0.00 and 0.10), so most of Yi’s +39.0pp comes from the full-KV fallback firing on tasks where plain SnapKV collapses. That outcome is accuracy-preserving but it is *not* evidence that the raw threshold transfers to Yi; the drop-vs- ρ prediction is inverted there. The z-scored variant of Section 5 resolves this: with a per-model standardization from an unlabeled pilot, Yi compresses for real (kept-KV 0.22–0.32) while still gaining $\Delta = +0.145$, and Llama improves to $\Delta = +0.174$. We state the fixed- τ failure plainly in Section 8.

7.7 Scaling formula verification

Empirical verification of Eq. (6). Table 13 reports the headroom $1 - A_{\text{full}}$, the observed ρ , and the ratio. On dilution-prone tasks the ratio clusters in $[0.25, 0.55]$. On NIAH-MK3 it is ≤ 0.02 regardless of headroom; the partition is visible in the ratio alone. These are in-range (4K/16K) statements; Section 7.4 reports a pre-registered out-of-range test at 32K, where the ordering survives but the dilution-side ratio grows beyond the in-range band.

Table 13: Verification of the scaling formula. The ratio $\rho/(1 - A_{\text{full}})$ is approximately constant on dilution-prone tasks across (model, context) cells and near zero on NIAH-MK3 regardless of headroom.

Sweep	$1 - A_{\text{full}}$	ρ	ratio
Qwen2.5-1.5B 4K VT	0.18	0.06	0.33
Qwen2.5-1.5B 16K VT	0.75	0.19	0.25
Qwen2.5-3B 4K VT	0.00	0.00	–
Qwen2.5-3B 16K VT	0.09	0.05	0.55
Qwen2.5-3B 4K FWE	0.24	0.01	0.04
Qwen2.5-3B 16K FWE	0.62	0.32	0.52
Qwen2.5-1.5B 4K NIAH-MK3	0.35	0.01	0.03
Qwen2.5-1.5B 16K NIAH-MK3	0.74	0.00	0.00

7.8 Single-task showcase: Mistral 4K NIAH-MK3

The strongest single-task win. $N = 100$. Plain SnapKV at $b = 0.0625$ collapses from 0.99 full-cache to 0.00; gated SnapKV holds 0.89 across all budgets. The gate is closed on 90/100 inputs (correct prediction of capacity-bound) with 10/100 false positives (gate opens, eviction destroys the needle).

7.9 Per-layer ablation

The default gating uses a single key-mask shared across layers. A per-layer variant (each layer derives its own keep mask from its own attention) improves the aggregation tasks at 4K (FWE by $1.8\times \rho$, `niah_multivalue` by $2.1\times \rho$ on Qwen2.5-1.5B 4K) and matches at 16K. Numbers in Appendix B.

Table 14: End-to-end efficiency of gated eviction (RULER 4K, batch 1, bf16, greedy, 128 decode steps, medians; A100-80GB). “Gated expected” is the gate-open-weighted mix $0.75 \cdot (b=0.125) + 0.25 \cdot \text{full}$ at the measured mixed-suite gate-open fraction ($\tau=0.07$); “gated measured” actually gates each input (15/20 open, matching the $N=400$ calibration fraction of 0.75). On the 1.5B model (2 KV heads), 4K decode is weight-bound and eviction buys *memory* ($8\times$ plain, $2.9\times$ gated expected), not speed; on Mistral-7B (8 KV heads, 513 MiB KV) eviction already yields a $1.29\times$ decode speedup at 4K. Measured KV matches the analytic $L \cdot H_{\text{KV}} \cdot d \cdot T \cdot 4 \text{ B}$ formula to 0.1 MiB, validating the 16K/32K scaling rows.

Model	Config	Decode tok/s	KV cache (MiB)	Peak decode (MiB)
Qwen2.5-1.5B ($N=20$)	Full KV	37.9	106.1	3090
	SnapKV $b=0.125$	37.8	13.2 ($8.0\times$)	2976
	SnapKV $b=0.0625$	37.5	6.6 ($16.1\times$)	2968
	Gated (measured, $\tau=0.07$)	37.7	13.2 / 110.5 (open/closed)	2976
	Gated (expected mix)	37.9	36.5 ($2.9\times$)	3004
Mistral-7B-v0.3 ($N=12$)	Full KV	18.9	512.8	14429
	SnapKV $b=0.125$	24.5 ($1.29\times$)	64.1 ($8.0\times$)	13929
Prefill-time overhead (Qwen2.5-1.5B, 4K, median ms)		Analytic KV at $b=0.125$ gated (0.75 open)		
Prefill (eager)		344	Qwen2.5-1.5B, 16K / 32K	448 $\rightarrow$ 154 / 896 $\rightarrow$ 308 MiB
SnapKV scoring		1.2	Mistral-7B, 16K / 32K	2.0 $\rightarrow$ 0.69 / 4.0 $\rightarrow$ 1.38 GiB
Head-agreement gate		17.2	(gate on Mistral-7B: 88.5 ms, $O(LH^2k)$ head-pair growth)	
Eviction + cache rebuild		3.2		

7.10 Random-eviction control

Uniform-random scoring at every budget destroys accuracy uniformly on both dilution-prone and capacity-bound inputs. The dilution effect is *not* “any eviction is regularization.” The control rules out a generic regularization explanation; the controlled numbers are in Appendix A.

7.11 Cost of the gate, and what the deployment delivers

The gate adds one prefill-time computation: $L \cdot H^2 \cdot k$ top- k Jaccard comparisons after the existing SnapKV scoring pass. Measured (Table 14): 17ms on Qwen2.5-1.5B at 4K ($\sim 5\%$ of the 344ms prefill, $< 1\%$ of end-to-end with 128 decode steps) and 89ms on Mistral-7B, whose $8\times$ larger head-pair count dominates the $O(LH^2k)$ term. The gate is free when the base evictor already runs the scoring pass; otherwise it shares the cost.

End-to-end, the payoff is scale-dependent and we report it honestly. On the 1.5B model at 4K, decode is weight-bound (2 KV heads, 106MiB cache): eviction buys *memory* ($8\times$ plain, $2.9\times$ under the gated mix at the measured 0.75 gate-open fraction), not speed. On Mistral-7B (8 KV heads, 513MiB cache), plain eviction at $b = 0.125$ already yields a $1.29\times$ decode speedup at 4K ($1.22\times$ under the gated mix), and the analytic KV formula (validated against measurement to 0.1MiB) puts the gated saving at $\sim 2.7\text{GiB}$ per sequence at 32K. Timings were taken on a shared A100 with a stable 25–29% background load, so throughputs are mild lower bounds; all configurations shared identical conditions.

7.12 Head-to-head with CapKV on LongBench

CapKV [Yang et al., 2026a] is the closest published method that reports on LongBench. No public CapKV codebase exists yet (their preprint is dated 2026-04-28); we re-implement CapKV’s per-(layer, kv-head) statistical-leverage score $s_i = w_i \cdot v_i^\top A^{-1} v_i$ with $A = I + \sum_i w_i v_i v_i^\top$ and $w_i = \exp(\langle k_i, \mu_q \rangle \beta / \sqrt{d})$ from their equations, then wrap it in our gate. Two LongBench subtasks, Qwen2.5-3B-Instruct, budget $b = 0.5$, CapKV temperature $\beta = 5$ (their default; we write β for their τ to avoid clashing with our gate threshold). Results

in Table 15.

Table 15: Gated vs. plain CapKV (our re-implementation) on two LongBench subtasks, Qwen2.5-3B-Instruct, $b = 0.5$.

Subtask	N	plain CapKV	gated CapKV	Δ
LongBench/qasper	68	0.132	0.147	+0.015
LongBench/triviaqa	24	0.875	0.958	+0.083

Both columns are positive at matched budget. Conditional on the gate firing (drop ≥ 0.07 , the same $\tau = 0.07$ we use everywhere), plain and gated agree by construction; on the gate-closed subset where the wrapper falls back to full KV, the difference is +4.3pp on qasper ($N_{\text{closed}} = 23$) and +18.2pp on triviaqa ($N_{\text{closed}} = 11$). The headline numbers are not a reproduction of CapKV’s Table 1 (different model, different metric, 16K eager cap on prompt length) and we report them as a directional head-to-head: at these sample sizes ($N = 68, 24$) wrapping CapKV in the gate is weakly positive overall and concentrates its gain on the gate-closed subset, consistent with the orthogonality claim of Section 4.2. We do not promote “never hurts” to a formal claim at these sample sizes; larger LongBench runs are clean follow-up work.

7.13 Head-to-head with DBTrimKV on RULER

Bui et al. [2026] (DBTrimKV) is the published trained-gate evictor the paper’s mechanism section adopts. The authors ship pretrained gates on Hugging Face for several Qwen3 variants (no Qwen2.5 checkpoint exists, so the head-to-head is necessarily on a different model family). We run DBTrimKV-Qwen3-4B-Instruct-2507 on RULER 4K, $N = 30$ inputs per task on (NIAH-MK3, FWE, VT, QA.1), at three memory budgets $M \in \{128, 256, 512\}$. The wrapper uses the same fixed $\tau = 0.07$ calibrated on Qwen2.5-1.5B; no per-model tuning. Results in Table 16.

Table 16: Gated vs. plain DBTrimKV (pretrained Qwen3-4B-Instruct-2507 gates) on RULER 4K at three memory budgets. See the caveat paragraph: the gate fires on only 1.7% of records, so the win is dominated by the full-KV fallback.

	N	plain DBTrimKV	gated DBTrimKV	Δ
all (3 budgets $\times$ 4 tasks)	360	0.617	0.850	+0.233
per task: NIAH-MK3	90	0.467	1.000	+0.533
per task: FWE	90	0.222	0.600	+0.378
per task: VT	90	0.989	1.000	+0.011
per task: QA.1	90	0.789	0.800	+0.011
per budget: $M = 128$	120	0.475	0.850	+0.375
per budget: $M = 256$	120	0.617	0.850	+0.233
per budget: $M = 512$	120	0.758	0.850	+0.092

Honest caveat: the gate fires rarely on Qwen3 at $\tau = 0.07$. At our default threshold the gate is open on $6/360 = 1.7\%$ of records. The remaining $\sim 98\%$ trigger the full-KV fallback, which explains the large Δ on NIAH-MK3 (+0.533): the gate closing turns off DBTrimKV’s compress step entirely. As a consequence, cache occupancy under our wrapper is $\sim 14.5\times$ plain DBTrimKV at the same nominal M . The +0.233 is therefore a *quality* win at matched budget, not a *memory-budget* win at matched cache size. The rare gate firing here is the same Qwen3 scale mismatch analyzed in Section 5: Qwen3-4B’s drop distribution ($\mu = 0.034$) puts the fixed $\tau = 0.07$ at $z = +2.0$, so a per-model z-scored threshold ($\tau_{\text{Qwen3}} = 0.022$) is the right deployment default and, applied to the same suite, restores a genuinely compressing gate (43% kept-KV, $\Delta = +0.235$). The headline is that wrapping the strongest published trained-gate evictor with our partition-

aware gate strictly improves matched-budget accuracy, with the gain concentrated on the capacity-bound task as the mechanism predicts.

8 Limitations

We list the limitations of the present work in roughly decreasing order of severity.

The sufficient-condition theorem is sketched, not formal. The explicit bridge from the head-agreement drop D to the dilution δ_t of Bui et al. [2026] is sketched in Appendix E. The sketch gives explicit forms for the constants c_1, c_2 on the simplified single-head surrogate but does not claim they are tight, nor does it rigorously prove the converse: the operational “capacity-bound $\Rightarrow$ small D ” direction requires a margin condition we have not verified on real attention. The fallback is the partition (Section 3) and the orthogonality demonstration (Section 4.2), neither of which depends on the bridge.

The fixed τ does not transfer to the Llama family. On Yi-1.5-9B and Llama-3.1-8B, `niah_multivalue` drops below $\tau = 0.07$ on both models; on Yi the gate additionally stays closed on FWE and VT (gate-open fractions 0.00 and 0.10), and on Llama the capacity-bound anchor NIAH-MK3 has mean $D = 0.073$, marginally *above* τ , rescued only by per-input dispersion. The accuracy deltas remain positive on both models, but on Yi that is because the conservative full-KV fallback happens to fire where plain SnapKV collapses — the drop-vs- ρ prediction itself is inverted there. What transfers to the Llama family is the ordering, not the raw threshold. The z-scored variant (Section 5) repairs this with an unlabeled per-model pilot — Llama Δ rises to +0.174 and Yi compresses genuinely at $\Delta = +0.145$ — but it adds a calibration step, its task-level claim excludes `niah_multivalue`, and per-input separation on the Llama family remains imperfect (AUC ≈ 0.80). A per-input statistic beyond mean D is the natural follow-up.

The capacity-bound claim is scoped to attention-score-based evictors. All four matrix base evictors (and the partition sweeps) score positions by attention statistics. ManifoldKV [Datta et al., 2026] reports that a Euclidean-outlier key scorer substantially improves NIAH-MultiKey retention at 50% compression, so “eviction is monotone damaging on NIAH-MK3” should be read as a statement about attention-score-based eviction, not about every possible scorer. Our single-mask geometric variant (Section 4.2) did not reproduce their rescue and the gate composes with it cleanly, but their exact per-layer, per-head design remains untested in our harness.

Single-seed runs; some cells are low-powered. All sweeps use greedy decoding and a single fixed seed; per-cell N ranges from 30 to 100 inputs (per-task N noted in each table). All headline deltas carry cluster-bootstrap 95% intervals and all ρ values Wilson intervals (Section 7, Appendix F); every matrix delta excludes zero, but the per-cell partition calls are individually low-powered (most straddle the 0.05 threshold), and small- N cells (Qwen-14B 16K, $N = 30$; triviaqa head-to-head, $N = 24$) remain noisy. The partition claim rests on the pooled contrast, which is unambiguous.

Qwen2.5-3B 16K needs τ -calibration. The fixed $\tau = 0.07$ default fails on this cell. The 40-input recipe of Section 5 fixes it (Δ goes from +3.4pp to +5.4pp). We recommend the recipe as the deployment default; we keep the fixed- τ result throughout to show the gap is small.

LongBench-SnapKV shows $\Delta \approx 0$ on the dilution-prone subset. On LongBench [Bai et al., 2024] qasper, triviaqa, trec, and multifoldqa_en under SnapKV scoring (plus passage_retrieval_en on the 1.5B model), the pooled gated Δ is within ± 1 pp of plain on all four models (Qwen2.5-1.5B 0.000, Qwen2.5-3B +0.002, Mistral-7B +0.003, Qwen2.5-14B -0.008). The worst per-task cell is Qwen2.5-14B trec at -2.2pp, the only negative gating result in the project, which we attribute to gate misfires on a task with no capacity-bound inputs to rescue. LongBench, on this subset, has no capacity-bound task; the partition predicts

neutrality and we observe it, including its mildly negative tail. The complementary capacity-bound-minority behaviour is documented in Section 7.12.

DBTrimKV head-to-head uses Qwen3 not Qwen2.5. DBTrimKV’s pretrained gates exist only for Qwen3 variants, so Section 7.13 runs on Qwen3-4B-Instruct-2507. The matrix is Qwen2.5-keyed, so DBTrimKV stays a standalone head-to-head rather than a matrix row.

4 base evictors, not 6. The matrix covers training-free policies. Trained / learned evictors (DBTrimKV, CapKV, IndexMem, DynamicKV) are cited but not in the matrix for the codebase-compat reason above.

The confirmed capacity-bound exemplar is RULER-only. The partition is verified on RULER 4K/16K across six task families plus two LongBench probes. NIAH-MK3 (and marginally MK2) remain the only tasks where capacity-bound failure is *empirically confirmed*. We looked for a non-RULER second exemplar: LongBench passage_retrieval_en turned out to be evict-robust (correctly gated open; Section 3), and LongBench passage_count is *gate-predicted* capacity-bound (mean $D = 0.054 < \tau$, gate closed on 43/47 inputs) but untestable at this model scale because full-KV accuracy is at floor (0.04), so there is nothing for eviction to destroy — consistent with the scaling formula, but not confirmation. Verifying passage_count on a stronger model is the natural test. Multilingual and code tasks are out of scope; we expect the predictor to transfer because the mechanism is architecture-level, not English-specific, but we have not measured it.

9 Conclusion

The benefit of KV-cache eviction is not uniform across inputs; it is sharply partitioned. The early-to-late head-agreement drop is a $\sim 50\text{ms}$ label-free statistic that predicts which side an input falls on, and a gating wrapper turns the prediction into a training-free policy: keep the full cache when the drop is small, run the base evictor when it is large. The wrapper is positive on every (model, base evictor, context) cell tested with a single threshold $\tau = 0.07$, and the worst single-task failure of plain SnapKV we observed (Mistral-7B NIAH-MK3, $99\% \rightarrow 0\%$ as budget shrinks) becomes a flat 89% under the gate.

Two questions remain open and we name them so the next paper does not re-derive them. First, the bridge from the head-agreement drop to the dilution δ_t is sketched in Appendix E on a single-head surrogate; the constants and the converse direction on real multi-head attention are not formally proven. Second, the Llama-architecture family mis-classifies niah_multivalue at our fixed τ , suggesting that the early-vs-late layer-bin choice may need to be model-family-specific for a fully cross-architecture deployment. Both have natural attack paths laid out in the gap list of Appendix E and the limitations of Section 8.

The mechanism is inherited from Bui et al. [2026]; the partition, the predictor, the wrapper, and the joint (M, T) scaling formula are new.

Reproducibility statement

All experiments run on single NVIDIA A100 80GB GPUs with PyTorch 2.11, Transformers 5.9, and Python 3.13, in bfloat16 with greedy (argmax) decoding, `max_new_tokens = 128`, and a fixed seed. Models are the public Hugging Face checkpoints of Qwen2.5-1.5B/3B/14B-Instruct, Mistral-7B-Instruct-v0.3, Yi-1.5-9B-Chat, and Llama-3.1-8B-Instruct; tasks come from the `simonjegou/ruler` port of RULER and from LongBench. Cache slicing preserves RoPE alignment via explicit `position_ids`; 16K runs use SDPA prefill with an eager re-forward of the $w = 32$ observation window for scoring (Appendix G). Per-cell sample sizes are stated in each table caption. The gate is one thresholded scalar (Eq. (3)), so every gated number in the paper can be recomputed from the released per-input logs; code, sweep configurations, and raw per-input jsonl outputs will be released with the camera-ready.

Ethics statement

This work studies inference-time efficiency of publicly released language models on public benchmarks. It involves no human subjects, no personal data, and no new model training. Better KV-cache eviction lowers the memory and energy cost of long-context inference; we see no ethical risk specific to this method beyond those of the underlying models.

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A Random-eviction control

We replace SnapKV scoring with uniform-random per-position scoring at every budget on the 4K, 4-key synthetic NIAH probe ($N = 150$, Qwen2.5-1.5B). The score-driven sweep retains accuracy at moderate budgets; the uniform-random sweep collapses immediately.

Table 17: SnapKV vs. uniform-random scoring at matched budget on the 4K 4-key synthetic NIAH probe. Random eviction destroys accuracy long before the budget gets aggressive; the dilution effect requires informed scoring.

budget b	SnapKV acc	Random acc	Δ
1.000	0.827	0.827	+0.000
0.875	0.820	0.400	+0.420
0.750	0.820	0.173	+0.647
0.625	0.827	0.040	+0.787
0.500	0.833	0.033	+0.800
0.375	0.807	0.007	+0.800
0.250	0.760	0.000	+0.760
0.125	0.207	0.000	+0.207
0.0625	0.033	0.000	+0.033

The per-input recovery rate ρ under random scoring is 0.027 (vs. 0.013 under SnapKV); the four “recovered” inputs under random are lucky hits among an otherwise destroyed distribution. The control rules out a generic regularisation explanation: any-eviction-helps is falsified.

B Per-layer eviction ablation

The default gating uses a single key-mask shared across layers. A per-layer variant derives each layer’s keep mask from that layer’s own last-window attention with the same total budget per layer (uniform cache shape). Table 18 reports ρ on the four dilution-prone RULER tasks where per-layer might plausibly help, on Qwen2.5-1.5B at 4K and 16K.

C Per-input recovery distributions

The mean ρ hides the shape of the per-input recovery, so we also report its distribution. For each input x we form the recovery indicator $r(x) = \text{sign}(\max_b A_{\text{gated}}(b, x) - A_{\text{plain}}(1, x)) \in \{-1, 0, +1\}$, where the maximum is over the budget sweep and $A_{\text{plain}}(1, x)$ is the full-cache plain accuracy. Figure 6 pools all (M, T) cells per task; Figure 7 reports the fraction of strictly recovered inputs ($r(x) > 0$) within each (M, T) cell.

Table 18: Per-layer vs. single-mask SnapKV on Qwen2.5-1.5B. Per-layer wins clearly on aggregation tasks at 4K (FWE 1.8 $\times$, niah_multivalue 2.1 $\times$); ties at 16K and on sequential-integration tasks.

Task	Context	Single-mask ρ	Per-layer ρ	Ratio
VT	4K	0.060	0.040	0.67
FWE	4K	0.060	0.110	1.83
QA.1	4K	0.080	0.070	0.88
niah_multivalue	4K	0.070	0.150	2.14
VT	16K	0.190	0.190	1.00
FWE	16K	0.040	0.050	1.25
QA.1	16K	0.140	0.140	1.00
niah_multivalue	16K	0.100	0.100	1.00

The histograms show that ρ is not driven by a few outliers: across the three dilution-prone tasks we recover a few percent to over ten percent of inputs in many cells, with no input ever made strictly worse by gating ($r=-1$ bin is empty). NIAH-MK3 sits at $r=0$ on essentially every input, confirming that the capacity-bound regime is a genuine null rather than a noisy average. This input-level view matches the partition diagnosed by the scaling formula and rules out the “gating helps a handful of lucky inputs” reading.

D Proof sketch of the scaling proposition

We expand the sketch from Section 6.

Setup. Let $T = R + N$ with R relevant and N noise positions. Assume noise isotropy: $\alpha_t \approx (1 - \alpha_{\mathcal{R}})/N$ for $t \in \mathcal{N}$, with $\alpha_{\mathcal{R}} := \sum_{t \in \mathcal{R}} \alpha_t$. Let σ_v^2 be the per-distractor value-vector variance under isotropy (distinct from the pre-softmax logit noise scale σ_z of App. E). The pre-eviction head SNR is

$$\text{SNR}_{\text{pre}} = \frac{\alpha_{\mathcal{R}}^2}{\sigma_v^2 \sum_{t \in \mathcal{N}} \alpha_t^2} \approx \frac{\alpha_{\mathcal{R}}^2 N}{\sigma_v^2 (1 - \alpha_{\mathcal{R}})^2}.$$

Eviction with noise-retention ratio $\gamma \in (0, 1)$ keeps a fraction $\rho_{\mathcal{R}}$ of the relevant mass and a fraction $\rho_{\mathcal{N}}$ of the noise mass. After softmax renormalisation over the kept set, the post-eviction relevant mass is $\tilde{\alpha}_{\mathcal{R}} = \rho_{\mathcal{R}} \alpha_{\mathcal{R}} / Z$ with $Z = \rho_{\mathcal{R}} \alpha_{\mathcal{R}} + \rho_{\mathcal{N}} (1 - \alpha_{\mathcal{R}})$, and the retained noise positions number γN with per-position mass $\rho_{\mathcal{N}} (1 - \alpha_{\mathcal{R}}) / (Z \gamma N)$. Substituting,

$$\text{SNR}_{\text{post}} = \frac{\tilde{\alpha}_{\mathcal{R}}^2}{\sigma_v^2 \sum_{t \in \mathcal{K} \cap \mathcal{N}} \tilde{\alpha}_t^2} \approx \frac{\rho_{\mathcal{R}}^2 \alpha_{\mathcal{R}}^2 \gamma N}{\sigma_v^2 \rho_{\mathcal{N}}^2 (1 - \alpha_{\mathcal{R}})^2} = \frac{\rho_{\mathcal{R}}^2}{\rho_{\mathcal{N}}^2 \gamma^{-1}} \cdot \text{SNR}_{\text{pre}}. \quad (8)$$

Under good scoring ($\rho_{\mathcal{R}} \geq \rho_{\mathcal{N}}$ with $\rho_{\mathcal{N}} \approx \gamma$ on a calibrated scorer), the gain factor reduces to $\rho_{\mathcal{R}}^2 / \gamma$, so the post-eviction SNR strictly exceeds the pre-eviction SNR. The renormalisation Z is the step that turns the raw fractions ($\rho_{\mathcal{R}}, \rho_{\mathcal{N}}$) into a usable SNR ratio; without it the argument would collapse to $\rho_{\mathcal{R}} / \gamma$ absent the squaring.

Factorisation. We work inside the surrogate, where the true accuracy indicator $\mathbf{1}[A(b, x) > A(b_{\max}, x)]$ is replaced by the surrogate indicator $\mathbf{1}[\hat{A}(\text{SNR}(b, x)) > \hat{A}(\text{SNR}(b_{\max}, x))]$ with $\hat{A} : [0, \infty) \rightarrow [0, 1]$ strictly increasing and saturating at 1. This replacement is the surrogate’s content: it asserts that accuracy is a deterministic function of head SNR. Within the surrogate the improvement event factorises as

$$\mathbf{1}[\hat{A}(\text{SNR}_{\text{post}}) > \hat{A}(\text{SNR}(b_{\max}, x))] = \mathbf{1}[\hat{A}(\text{SNR}(b_{\max}, x)) < 1] \cdot \mathbf{1}[\hat{A}(\text{SNR}_{\text{post}}) > \hat{A}(\text{SNR}_{\text{pre}})].$$

Both monotonicity and saturation are needed: if $\hat{A}$ were not strictly increasing the second indicator could fire without an SNR gain, and if $\hat{A}$ did not saturate at 1 the case $\hat{A}(\text{SNR}(b_{\max}, x)) = 1$ would not zero out

the left-hand side. The first indicator is the headroom in the surrogate (a population-level proxy for the true full-cache failure probability $1 - A_{\text{full}}$); the second depends only on $\rho_{\mathcal{R}}/\gamma$ via Eq. (8) once $\rho_{\mathcal{N}} \approx \gamma$ is folded in. The transfer from surrogate event to operational accuracy is mediated by the surrogate-fits-data assumption (we treat $\hat{A}$ as a good fit to the expected accuracy conditional on SNR; deviations are absorbed into the empirical constant in $[0.25, 0.55]$). Marginalising over the prompt distribution and treating the two indicators as independent gives

$$\rho_{\text{KV}}(\mathbf{T}, M, T) \approx (1 - A_{\text{full}}(M, T)) \cdot p_{\text{recov}}(\rho_{\mathcal{R}}/\gamma),$$

which is Eq. (6). The boundary conditions on p_{recov} follow from the surrogate: at $\rho_{\mathcal{R}} = 0$ no relevant mass survives, so $\text{SNR}_{\text{post}} = 0$ and recovery is impossible, giving $p_{\text{recov}}(0) = 0$; at $\gamma = 1$ (no eviction) $\text{SNR}_{\text{post}} = \text{SNR}_{\text{pre}}$ identically, so the second indicator never fires and $p_{\text{recov}}(1) = 0$.

From surrogate to operational ρ_{KV} . The factorisation argues about the improvement event at a single budget b , whereas the operational ρ_{KV} (Eq. (1)) is the per-input strict-Pareto frequency over the best budget $b^*(x)$. The two are related by a union bound over the discrete budget grid $\mathcal{B} = \{b_1, \dots, b_K\}$:

$$\Pr_x[A(b^*(x), x) > A(b_{\text{max}}, x)] = \Pr_x[\exists b \in \mathcal{B} : A(b, x) > A(b_{\text{max}}, x)] \leq \sum_{b \in \mathcal{B}} \Pr_x[A(b, x) > A(b_{\text{max}}, x)].$$

Under monotone recovery (p_{recov} non-decreasing in $\rho_{\mathcal{R}}/\gamma$, with $\rho_{\mathcal{R}}$ determined by the scorer’s top- k at each b), the right-hand side is bounded by $K \cdot \Pr_x[A(\tilde{b}, x) > A(b_{\text{max}}, x)]$ for the budget $\tilde{b}$ that maximises the per-budget improvement probability. Under the independence approximation of the previous paragraph, this maximiser term equals Eq. (6), so

$$\rho_{\text{KV}}(\mathbf{T}, M, T) \leq K \cdot (1 - A_{\text{full}}(M, T)) \cdot p_{\text{recov}}(\rho_{\mathcal{R}}/\gamma).$$

This is a one-sided upper-bound at the cost of a multiplicative grid factor K (in our experiments $K \leq 7$, which is absorbed into the unknown empirical constant in $[0.25, 0.55]$ reported below). We use it to justify reading Eq. (6) as a population-level surrogate for ρ_{KV} up to a budget-grid constant.

Independence bound via covariance. Let A and B denote the headroom and recovery indicators viewed as Bernoulli random variables under the prompt-distribution measure. The exact decomposition is $\Pr[A \cdot B] = \Pr[A] \Pr[B] + \text{Cov}(A, B)$, so the independence step incurs only the covariance residual. Both indicators are monotonically decreasing in $\alpha_{\mathcal{R}}$: larger $\alpha_{\mathcal{R}}$ raises SNR_{pre} , pushing A_{full} toward 1 (less headroom, so A trends to 0) and simultaneously saturating SNR_{post} earlier under (A1) noise-isotropy (less recovery margin, so B trends to 0). Coupling A and B through the single scalar $\alpha_{\mathcal{R}}$ makes them comonotone in this driver, so the FKG inequality gives $\text{Cov}(A, B) \geq 0$, i.e. $\Pr[A \cdot B] \geq \Pr[A] \Pr[B]$. The Cauchy–Schwarz / Bernoulli-variance bound $|\text{Cov}(A, B)| \leq \sqrt{\text{Var}(A) \text{Var}(B)} \leq 1/4$ then yields an absolute, distribution-free residual. Combining with the union-bound paragraph above,

$$\rho_{\text{KV}}(\mathbf{T}, M, T) \leq K \cdot (1 - A_{\text{full}}(M, T)) \cdot p_{\text{recov}}(\rho_{\mathcal{R}}/\gamma) + K/4.$$

The $K/4$ term is loose but rigorous, replacing the prior leading-order hand-wave by a one-sided inequality that survives any prompt distribution. With $K \leq 7$ on our grid the absolute residual is at most 1.75, while the empirical ratios in Table 13 cluster in $[0.25, 0.55]$, consistent with the dominant $K \cdot (1 - A_{\text{full}}) \cdot p_{\text{recov}}$ term carrying the explanatory load and the covariance slack staying well inside the bound.

E Proof sketch: sufficient condition for the partition

We sketch a proof of the bridge inequality (7): under the simplified attention model below, a large head-agreement drop D implies a non-trivial lower bound on Bui’s dilution δ_t [Bui et al., 2026]. The argument is structured as five numbered steps followed by a converse heuristic, an empirical-validation paragraph, and an explicit list of what is *not* proved. We give closed forms for the constants c_1 and $c_2(\gamma)$ rather than leave them implicit. The constants carry an $O(H)$ multiplicative slack from the pairwise-to- H -wise Jaccard step in Step 3; tightening the constants is left to future work.

Notation reminder. $T = R + N$ key positions partitioned into the relevant set $\mathcal{R}$ ($|\mathcal{R}| = R$) and noise set $\mathcal{N}$ ($|\mathcal{N}| = N$), with $R \ll N$. There are L layers and H heads. For layer $\ell \in \{1, \dots, L\}$ and head $h \in \{1, \dots, H\}$, the pre-softmax logits are $z_{h,i}^{(\ell)} = (q_h^{(\ell)})^\top k_i + \eta_{h,i}^{(\ell)}$, where $\eta_{h,i}^{(\ell)}$ is zero-mean σ_z -sub-Gaussian noise independent across (ℓ, h, i) . The subscript z distinguishes the pre-softmax logit noise scale used here from the value-vector variance σ_v used in Appendix D. We write $\alpha_{h,i}^{(\ell)}$ for the post-softmax attention. Bui’s dilution $\delta_t = 1 - \sum_{i \in U_t} \alpha_{t,i}$ is taken at the decoding query at step t ; we drop the t subscript and write $\delta := \delta_t$.

Step 1 (the surrogate model). We use the multi-head lift of the single-head surrogate from Appendix D:

- (A1) *Signal coherence.* For each layer ℓ and head h , there is a head-private relevant subset $\mathcal{R}_h^{(\ell)} \subseteq \mathcal{R}$ such that for $i \in \mathcal{R}_h^{(\ell)}$ the deterministic part of the logit satisfies $(q_h^{(\ell)})^\top k_i \geq m_+$, and for $i \in \mathcal{N}$, $(q_h^{(\ell)})^\top k_i \leq m_-$, with $m_+ - m_- \geq m > 0$. This m is the *logit margin*; it is bounded above by $\log T$ in the well-trained regime.
- (A2) *Sub-Gaussian noise.* $\eta_{h,i}^{(\ell)}$ is zero-mean and σ_z^2 -sub-Gaussian, independent across (ℓ, h, i) . This is the standard noise model used in attention-as-feature-aggregation analyses.
- (A3) *Capacity-bound vs dilution-prone regimes.* In the capacity-bound regime $\mathcal{C}$, all heads at all layers share a common relevant set: $\mathcal{R}_h^{(\ell)} = \mathcal{R}$ for every (ℓ, h) . In the dilution-prone regime $\mathcal{D}$, late-layer head subsets diverge: there exists a layer index $\ell^* > L/2$ such that for any head pair (h, h') at layer ℓ^* , $|\mathcal{R}_h^{(\ell^*)} \cap \mathcal{R}_{h'}^{(\ell^*)}| / |\mathcal{R}_h^{(\ell^*)} \cup \mathcal{R}_{h'}^{(\ell^*)}| \leq J_{\text{late}}$, while early-layer heads share their subsets, $|\mathcal{R}_h^{(1)} \cap \mathcal{R}_{h'}^{(1)}| / |\mathcal{R}_h^{(1)} \cup \mathcal{R}_{h'}^{(1)}| \geq J_{\text{early}}$, with $J_{\text{early}} > J_{\text{late}}$. Both J_{early} and J_{late} are model-specific population constants of the surrogate, not functions of L .
- (A4) *Identification of U_t .* We identify Bui’s useful set U_t (a decoding-step quantity defined at the residual-stream level) with the head-common signal set $\mathcal{I}^{(\ell^*)} := \bigcap_h \mathcal{R}_h^{(\ell^*)}$ at the divergence layer ℓ^* (a single-layer attention quantity). The identification is a modelling choice (positions every head agrees are signal); it collapses time-step and layer-index axes, and the two coincide in the surrogate by fiat. We flag it in the “What is not proved” list below.

The empirical content of (A3) is the layer-dependence we measure: early layers respond to local context (similar $\mathcal{R}_h^{(\ell)}$ across heads, regardless of task), late layers respond to task structure (head divergence on dilution-prone tasks, head convergence on capacity-bound tasks). See §4.1 for the corresponding operational definitions.

Step 2 (head-agreement drop as Jaccard divergence). The head-agreement drop D in Eq. (3) is the early-late difference of mean pairwise top- k Jaccard agreement, with k chosen to be commensurate with R (in the experiments $k = 32 \sim |\mathcal{R}|$). In the high-SNR limit of the surrogate (Lemma 1 below), each head’s top- k set concentrates on its own private $\mathcal{R}_h^{(\ell)}$ with high probability. Hence at layer ℓ ,

$$a_\ell \approx \frac{1}{\binom{H}{2}} \sum_{h < h'} \frac{|\mathcal{R}_h^{(\ell)} \cap \mathcal{R}_{h'}^{(\ell)}|}{|\mathcal{R}_h^{(\ell)} \cup \mathcal{R}_{h'}^{(\ell)}|}. \quad (9)$$

Binning into early third and late third and subtracting,

$$D = a_{\text{early}} - a_{\text{late}} \approx \bar{J}_{\text{early}} - \bar{J}_{\text{late}}, \quad (10)$$

the difference of mean pairwise Jaccard between layers in the early and late bins. D is therefore a direct measurement of the head-set divergence in (A3).

Lemma 1 (Top- k concentration on the private set). *Under (A1), (A2) with $m \geq \sigma_z \sqrt{4 \log(2H^2 L T^2 / \beta)}$, the top- k set of head (ℓ, h) equals $\mathcal{R}_h^{(\ell)}$ jointly across all HL heads with probability $\geq 1 - \beta$, when $k = |\mathcal{R}_h^{(\ell)}|$.*

Proof. A standard sub-Gaussian maximal inequality: for any pair (i, j) with $i \in \mathcal{R}_h^{(\ell)}$ and $j \in \mathcal{N}$, $\Pr[z_{h,i}^{(\ell)} < z_{h,j}^{(\ell)}] \leq \exp(-m^2/(4\sigma_z^2))$. There are $R \cdot N \leq T^2/4$ such relevant-vs-noise pairs at each layer-head. Substituting the threshold $m \geq \sigma_z \sqrt{4 \log(2H^2 LT^2/\beta)}$ gives a per-pair tail of $\beta/(2H^2 LT^2)$. Union-bounding over up to T^2 pairs per layer-head and over all HL layer-heads gives a total tail of $HL \cdot T^2 \cdot \beta/(2H^2 LT^2) = \beta/(2H) \leq \beta$. $\square$

The threshold $m \geq \sigma_z \sqrt{4 \log(2H^2 LT^2/\beta)}$ is the *margin condition*; if it holds, every step below holds with probability $\geq 1 - \beta$ over the noise.

Step 3 (head-set divergence forces union expansion). For each layer ℓ define the union of head-private sets $\mathcal{U}^{(\ell)} := \bigcup_{h=1}^H \mathcal{R}_h^{(\ell)}$ and the intersection $\mathcal{I}^{(\ell)} := \bigcap_{h=1}^H \mathcal{R}_h^{(\ell)}$. The mean pairwise Jaccard $\bar{J}_\ell$ at layer ℓ is a *pairwise* object, whereas $|\mathcal{U}^{(\ell)}|$ and $|\mathcal{I}^{(\ell)}|$ are H -wise objects; the ratio $|\mathcal{U}|/|\mathcal{I}|$ is not equal to $1 - \bar{J}$ in general. Inclusion-exclusion only gives $|\mathcal{U}|/|\mathcal{I}| \leq 1 + (H - 1)(1 - \bar{J})$ in the best case, and the union-vs-intersection gap can scale with H in the worst case. We therefore avoid an H -wise identity and use the weaker pairwise inequality. Under (A3), $\bar{J}_{\ell^*} \leq J_{\text{late}}$, so there exists a head pair (h, h') at layer ℓ^* with $|\mathcal{R}_h^{(\ell^*)} \cap \mathcal{R}_{h'}^{(\ell^*)}| \leq J_{\text{late}} \cdot |\mathcal{R}_h^{(\ell^*)} \cup \mathcal{R}_{h'}^{(\ell^*)}|$. Picking the head $h^\dagger$ with the largest private residual set $\mathcal{R}_{h^\dagger}^{(\ell^*)} \setminus \mathcal{I}^{(\ell^*)}$,

$$|\mathcal{U}^{(\ell^*)} \setminus \mathcal{I}^{(\ell^*)}| \geq |\mathcal{R}_{h^\dagger}^{(\ell^*)} \setminus \mathcal{I}^{(\ell^*)}| \geq (1 - J_{\text{late}}) \cdot \max_h |\mathcal{R}_h^{(\ell^*)}|. \quad (11)$$

This is strictly weaker than treating $(|\mathcal{U}| - |\mathcal{I}|)/|\mathcal{U}|$ as if it were equal to $1 - \bar{J}$, but it is correct; the slack is absorbed in c_1 below.

Under (A4), Bui's useful set is $U_t = \mathcal{I}^{(\ell^*)}$, so positions in $\mathcal{U}^{(\ell^*)} \setminus \mathcal{I}^{(\ell^*)}$ are attended by some head but not by all, and from the perspective of the next layer's query they contribute attention mass outside U_t .

Step 4 (bridge to Bui's δ_t). Let $\bar{\alpha}$ denote the average per-head, per-position attention on a position in $\mathcal{R}_h^{(\ell^*)}$ at layer ℓ^* . Under (A1) with margin m , the softmax concentrates and $\bar{\alpha} \geq (1 - e^{-m})/R$ on private positions. Treating per-head value contributions as additive after the output projection (a modelling choice we flag below), the attention mass *outside* $\mathcal{I}^{(\ell^*)}$ contributed by the maximal-residual head is at least

$$\sum_{i \in \mathcal{R}_{h^\dagger}^{(\ell^*)} \setminus \mathcal{I}^{(\ell^*)}} \alpha_{h^\dagger, i}^{(\ell^*)} \geq \bar{\alpha} \cdot (1 - J_{\text{late}}) \cdot \max_h |\mathcal{R}_h^{(\ell^*)}|, \quad (12)$$

using Eq. (11). Under eviction with noise-retention ratio γ and a top- k scorer that does not a priori distinguish signal-side residuals from noise positions (an additional modelling assumption beyond (A1)–(A4) which we make explicit here), positions in $\mathcal{U}^{(\ell^*)} \setminus \mathcal{I}^{(\ell^*)}$ inherit the same effective per-position retention rate γ as the noise set. Those retained for only a subset of heads appear as distractors to the remaining heads after the cache is merged. Conditional on this retention,

$$\mathbb{E}[\delta_t] \geq \bar{\alpha} \cdot (1 - J_{\text{late}}) \cdot \max_h |\mathcal{R}_h^{(\ell^*)}| \cdot (1 - \gamma), \quad (13)$$

because the $(1 - \gamma)$ fraction of un-retained positions in the residual private sets no longer balance the heads that did keep them, leaving a residual distractor mass proportional to $(1 - \gamma)$. This is a new bound derived in our surrogate; it is structurally analogous to Bui's Corollary 3.2 (which relates pre- and post-eviction dilution via $\delta' = (\rho_D/\rho_U)\delta/[(1 - \delta) + (\rho_D/\rho_U)\delta]$), but the input here is the head-disagreement-induced distractor set rather than a pre-existing dilution.

Step 5 (closing the bridge: solve for D). Eq. (13) uses $(1 - \bar{J}_{\text{late}})$, which we relate to the observable D via Eq. (10): $D = \bar{J}_{\text{early}} - \bar{J}_{\text{late}}$. Since $\bar{J}_{\text{early}} \leq 1$ trivially (Jaccard is bounded by one), $D \leq 1 - \bar{J}_{\text{late}}$, so substituting gives

$$\mathbb{E}[\delta_t] \geq \bar{\alpha} \cdot D \cdot \max_h |\mathcal{R}_h^{(\ell^*)}| \cdot (1 - \gamma). \quad (14)$$

The substitution $1 - \bar{J}_{\text{late}} \geq D$ holds always; we do not require the stronger $\bar{J}_{\text{early}} \approx 1$ assumption. The looser the value of $\bar{J}_{\text{early}}$, the larger the gap between $1 - \bar{J}_{\text{late}}$ and D , and the looser this bound is relative to its tightest form.

The $\log(1/\gamma)$ factor on the right-hand side of Eq. (7) is a heuristic amplification term, not a rigorous consequence of the steps above. The intended argument is that after eviction the cache retains γN noise positions, and the maximum of γN sub-Gaussian logits concentrates near $\sigma_z \sqrt{2 \log \gamma N}$; the multiplicative penalty on the head signal-to-noise budget should scale with the post-eviction noise floor relative to the pre-eviction floor. A careful Taylor expansion of $\sqrt{2 \log N} - \sqrt{2 \log \gamma N}$ gives a difference of order $\log(1/\gamma)/\sqrt{2 \log N}$, not $\log(1/\gamma)$ itself; the log factor therefore captures only the worst-case noise-tail shift at small N and not the asymptotic regime. We retain the $\log(1/\gamma)$ factor in Eq. (7) as a convenient packaging of the eviction amplification, but we flag this as gap (G5) in the list below. Readers willing to drop the factor entirely can read Eq. (7) as $D \geq c_1$ with $c_1 = 1/J_{\text{early}}$; this is the conservative no-amplification reading, and it is also empirically consistent with the $\tau = 0.07$ threshold we calibrate (Section 5).

Two heuristic observations close the bridge to $(1 - A_{\text{full}})$; both are leading-order estimates rather than rigorous bounds, and we flag them in the gap list. (i) Under (A1) noise-isotropy and the high-SNR regime, the average per-position mass on a relevant position satisfies $\bar{\alpha} \approx \alpha_{\mathcal{R}}/R$, and under (A3) the head-private sets collectively cover most of $\mathcal{R}$ in the dilution-prone regime so that $\max_h |\mathcal{R}_h^{(\ell^*)}| \approx R$. The product is therefore $\bar{\alpha} \cdot \max_h |\mathcal{R}_h^{(\ell^*)}| \approx \alpha_{\mathcal{R}}$ as an order-of-magnitude estimate. (The set containment $\mathcal{R}_h^{(\ell^*)} \subseteq \mathcal{R}$ alone gives the reverse inequality $\bar{\alpha} \cdot \max_h |\mathcal{R}_h^{(\ell^*)}| \leq \alpha_{\mathcal{R}}$, so the leading-order substitution is the honest reading and is flagged in the gap list.) (ii) Combining the headroom factorisation of Appendix D (Eq. (6)) with the strictly increasing accuracy surrogate gives $(1 - A_{\text{full}}) \leq C \cdot \alpha_{\mathcal{R}}$ on dilution-prone tasks for a constant C depending only on the slope of $\hat{A}(\cdot)$ near the operating SNR.

Combining (i) (as a leading-order estimate) and (ii) with Eq. (14) yields the operational bridge

$$\mathbb{E}[\delta_t] \gtrsim \frac{(1-\gamma)}{C} \cdot D \cdot (1 - A_{\text{full}}), \quad (15)$$

where $\gtrsim$ denotes leading-order in the surrogate; the substitution $\bar{\alpha} \cdot \max_h |\mathcal{R}_h^{(\ell^*)}| \approx \alpha_{\mathcal{R}}$ is the source of the $\gtrsim$ rather than $\geq$. The conservative rigorous reading replaces $\alpha_{\mathcal{R}}$ on the right-hand side with $\bar{\alpha} \cdot \max_h |\mathcal{R}_h^{(\ell^*)}| \leq \alpha_{\mathcal{R}}$ and drops the bridge to $(1 - A_{\text{full}})$ entirely; we use $\gtrsim$ for the operational reading.

Eq. (15) is the load-bearing bridge between the observable D and Bui’s dilution under the empirical headroom proxy. Eq. (7) in Section 6 packages this relationship as a threshold inequality. Substituting $D \geq T$ into Eq. (15) gives $\mathbb{E}[\delta_t] \gtrsim ((1-\gamma)T/C) \cdot (1 - A_{\text{full}})$, so any chosen threshold T on D produces a corresponding $c_2 = (1-\gamma)T/C$. Choosing $T = c_1 \log(1/\gamma)$ with $c_1 = 1/J_{\text{early}}$,

$$\boxed{c_1 = \frac{1}{J_{\text{early}}}, \quad c_2(\gamma) = \frac{(1-\gamma) \cdot \log(1/\gamma)}{C \cdot J_{\text{early}}}.} \quad (16)$$

J_{early} enters the threshold as a model-specific scale that makes c_1 commensurate with the empirical Jaccard floor across heads. On our test models $J_{\text{early}} \in [0.4, 0.7]$, giving $c_1 \in [1.4, 2.5]$; the empirical $\tau = 0.07$ is well below this, consistent with the constants being loose (per the $O(H)$ slack in Step 3 and the heuristic substitution in observation (i) above). Both constants carry an $O(H)$ multiplicative slack from the pairwise-to- H -wise step of Step 3, and the $\log(1/\gamma)$ factor in $c_1 \log(1/\gamma)$ is heuristic per gap (G5). The conservative no-amplification reading drops the $\log(1/\gamma)$ factor and gives the threshold $D \geq c_1 = 1/J_{\text{early}}$ with rate $c_2(\gamma) = (1-\gamma)/(C \cdot J_{\text{early}})$ directly.

The converse direction (heuristic, sufficient-side only). The partition predictor’s operational claim is the contrapositive: *small $D \Rightarrow$ low dilution $\Rightarrow$ eviction is risky*. We sketch two arguments and flag that neither is a separate necessity proof.

The first is a comment on the sufficient-side lower bound: under (A3) for the capacity-bound regime $\mathcal{C}$ ($\mathcal{R}_h^{(\ell)} = \mathcal{R}$ for all (ℓ, h)), the union and intersection coincide, so the right-hand side of Eq. (13) is zero. Since

Eq. (13) is a *lower* bound on $\mathbb{E}[\delta_t]$, a zero right-hand side does not bound δ_t from above; it only says the sufficient lower bound becomes vacuous.

The second is an upper bound on δ_t that does provide some necessary-side content, but only inside the surrogate. Under (A1) with margin m satisfying the threshold of Lemma 1, and (A3) regime $\mathcal{C}$ so that $U_t = \mathcal{R}$ for every head, the pairwise top- k Jaccard between heads is $\geq J_{\text{early}}$ at every layer (no late-layer divergence), so $|\mathcal{U}|/|\mathcal{I}| \rightarrow 1$ and there is no residual distractor mass. The attention mass on $\mathcal{N}$ for any single head is bounded by the sub-Gaussian tail $O(\exp(-m^2/(4\sigma_z^2)))$ by Lemma 1, and Bui’s dilution is $\delta_t \leq O(\exp(-m^2/(4\sigma_z^2)))$ in the surrogate. This is the surrogate-level converse; lifting it to real attention requires both a lower bound on m for capacity-bound tasks (which we have not verified) and the surrogate identification (A4) of U_t with $\mathcal{I}^{(\ell^*)}$. The boxed Eq. (7)/(16) is the sufficient direction only.

Empirical validation. Table 12 reports the mean head-agreement drop D across the five dilution-prone tasks (qa_1, qa_2, vt, niah_multivalued, fwe) and the capacity-bound task (NIAH-MK3) on Qwen 1.5B, Qwen 3B, and Mistral. In every (model, context) cell, NIAH-MK3 has the smallest drop (including the cell where it is negative, Qwen 3B 16K). The *ordering* of tasks by D is preserved across architectures, and the partition between dilution-prone and capacity-bound is exactly the partition predicted by Eq. (15): cells with $D \approx 0$ have $\mathbb{E}[\delta_t]$ bounded by the noise floor, and the per-input recovery rate ρ in Table 1 is ≤ 0.02 on all such cells. The bridge inequality is therefore empirically consistent with both directions on this task suite. We do not promote this to a formal claim because the sample size per cell is $N = 20$ inputs, the constants in Eq. (16) are not tight, and the necessity direction is not proved.

What is *not* proved. We list the gaps so a reviewer reading this appendix sees the scope clearly.

1. **The constants $c_1, c_2(\gamma)$ in Eq. (16) are loose.** The pairwise-to- H -wise step in Step 3 loses tightness: the union-vs-intersection gap can scale with H in the worst case, so c_1 is honest only up to an $O(H)$ multiplicative slack. A tight version would replace the pairwise inequality with a full higher-order union-intersection identity and track the layerwise covariance of head subsets.
2. **The pairwise Jaccard is not the same object as the H -wise union/intersection ratio.** Eq. (11) uses the weaker inequality “some head has a large private residual set”; this is correct but discards information when many heads share the same residual.
3. **Observation (i) in Step 5 is a leading-order estimate, not a rigorous inequality.** The exact reverse inequality $\bar{\alpha} \cdot \max_h |\mathcal{R}_h^{(\ell^*)}| \leq \alpha_{\mathcal{R}}$ holds (set containment), so the substitution $\bar{\alpha} \cdot \max_h |\mathcal{R}_h^{(\ell^*)}| \approx \alpha_{\mathcal{R}}$ used in Eq. (15) relies on (A1)+(A3) jointly to push the high-SNR approximation $\max_h |\mathcal{R}_h^{(\ell^*)}| \approx R$. Eq. (15) therefore uses $\gtrsim$ rather than $\geq$.
4. **The identification of Bui’s U_t with the head-common signal set $\mathcal{I}^{(\ell^*)}$ (A4) is a modelling choice, not a derived equality.** Bui’s U_t is defined at the residual-stream level; our $\mathcal{I}^{(\ell^*)}$ is an intersection over per-head attended sets at a single layer. The two coincide in the surrogate by fiat.
5. **The $\log(1/\gamma)$ factor in Step 5 is heuristic.** The sub-Gaussian maximum over γN retained-noise positions shifts noise-tail scale by $O(\log(1/\gamma)/\sqrt{\log N})$, not $\log(1/\gamma)$. We retain the cleaner $\log(1/\gamma)$ packaging in Eq. (7) as a worst-case bound at small N , and we offer the conservative reading $D \geq c_1$ (no log term) as a strict alternative for readers who reject the surrogate-level amplification argument.
6. **The converse direction is conditional on a margin assumption we have not verified.** In Step 5 we used $m \geq \sigma_z \sqrt{4 \log(2H^2 L T^2 / \beta)}$ to get Lemma 1; the surrogate-level converse additionally requires that capacity-bound tasks satisfy this margin condition for the *common* relevant set $\mathcal{R}$. We do not test this on real attention weights; doing so is a clean follow-up.
7. **The converse paragraph above is an upper bound on the sufficient-condition argument, not a separate necessary-condition proof.** Eq. (7) with constants from Eq. (16) is the sufficient direction only; the empirical content of necessity is reported but not formalised.

8. **The assumption that heads share a common $\mathcal{R}$ in capacity-bound tasks (A3, regime C) needs verification.** Our evidence is the small empirical D on NIAH-MK3 across cells; whether the underlying mechanism is actually “all heads agree on the same needle position” or some weaker form of agreement (heads agreeing on a length- $O(1)$ neighborhood of the needle) is open. The difference between these is invisible to top- k Jaccard but matters for the rigorous proof.
9. **Step 4 implicitly assumes the scorer does not distinguish signal-side residual positions from noise positions a priori.** The $(1 - \gamma)$ factor in Eq. (13) applies the noise-retention ratio γ to positions in $\mathcal{U}^{(\ell^*)} \setminus \mathcal{I}^{(\ell^*)} \subseteq \mathcal{R}$, which are signal-side. The retention rate there equals γ only when a top- k scorer is symmetric across signal vs noise on these residuals; under (A1) noise-isotropy this is the leading-order assumption.
10. **The output-projection averaging in Step 4 treats per-head value contributions as additive.** The single-head SNR analysis of Appendix D is lifted to multi-head by summing attention masses across heads. A precise lift would use the value-vector subspace decomposition of the output projection; under non-orthogonal value subspaces this can shift the constants by $O(H)$ in the worst case, not the $O(1)$ we previously claimed.
11. **The assumed accuracy function $\hat{A}(\text{SNR})$ is treated as task-agnostic.** The constant C in Eq. (15) depends on the slope of $\hat{A}$ near the operating SNR; on tasks where $\hat{A}$ is steep (precise retrieval) versus shallow (aggregation), C differs by an order of magnitude.

The sketch is therefore a *structural* proof of the sufficient direction: the form of the bound and the dependence on (H, γ) are derived from the surrogate; the numerical values of the constants, the H -tightness, the $\log(1/\gamma)$ factor, and the necessity direction are flagged as future work.

F Confidence intervals for the headline numbers

Tables 19 and 20 report 95% intervals for every headline delta, computed as described in the uncertainty paragraph of Section 7. Full per-cell derivations ship with the released logs.

G Reproduction notes

Implementation details that matter for exact reproduction. The cache slicing uses explicit `position_ids` to preserve RoPE alignment; the naive attention-mask trick produces a silent position shift on evicted positions. Long-context (16K) runs use SDPA for prefill and switch to eager for the $w \times T$ observation-window scoring pass to avoid the $O(T^2)$ attention-matrix allocation on 80GB. Head agreement uses Jaccard similarity of top-32 key sets per head pair, averaged over pairs, with early/late layer bins $[0, \lfloor L/3 \rfloor)$ and $[\lfloor 2L/3 \rfloor, L)$. Decoding is greedy throughout, so the gated outcome of any input is exactly reproducible from its recorded head-agreement drop plus the plain-eviction and full-KV outcomes; we use this identity to evaluate gate thresholds post-hoc, and one matrix cell (Qwen2.5-3B 4K SnapKV, executed at $\tau = 0.04$) is reported at $\tau = 0.07$ via this exact re-evaluation. A pipeline audit verifies that eviction at budget 1.0 matches clean prefill-decode token-for-token up to a trailing-period mismatch on one of six audited examples; the released logs include the audit script.

Table 19: 95% confidence intervals for the 16 gating deltas of Table 4. Cluster bootstrap over inputs: we resample the (task, id) pairs with replacement (10,000 replicates); each input contributes its per-budget paired gated–plain differences, preserving the pairing and the within-input correlation across budgets; intervals are 2.5–97.5 percentiles. All gated outcomes are the exact $\tau = 0.07$ post-hoc reconstruction of Appendix G. $N = 400$ inputs per cell (4 tasks $\times$ 100), $N = 200$ for Qwen2.5-14B. *Every one of the 16 intervals excludes zero*; the smallest lower bound is +0.093 (Qwen2.5-1.5B, SnapKV).

Model	Base evictor	Δ	95% CI	N
Qwen2.5-1.5B	SnapKV	+0.121	[+0.093, +0.151]	400
Qwen2.5-1.5B	H2O	+0.162	[+0.126, +0.199]	400
Qwen2.5-1.5B	StreamingLLM	+0.163	[+0.128, +0.200]	400
Qwen2.5-1.5B	PyramidKV	+0.152	[+0.118, +0.188]	400
Qwen2.5-3B	SnapKV	+0.259	[+0.225, +0.295]	400
Qwen2.5-3B	H2O	+0.303	[+0.261, +0.347]	400
Qwen2.5-3B	StreamingLLM	+0.430	[+0.383, +0.477]	400
Qwen2.5-3B	PyramidKV	+0.329	[+0.286, +0.372]	400
Qwen2.5-14B	SnapKV	+0.143	[+0.108, +0.181]	200
Qwen2.5-14B	H2O	+0.200	[+0.152, +0.250]	200
Qwen2.5-14B	StreamingLLM	+0.250	[+0.190, +0.310]	200
Qwen2.5-14B	PyramidKV	+0.228	[+0.175, +0.285]	200
Mistral-7B	SnapKV	+0.187	[+0.154, +0.220]	400
Mistral-7B	H2O	+0.236	[+0.196, +0.278]	400
Mistral-7B	StreamingLLM	+0.263	[+0.220, +0.305]	400
Mistral-7B	PyramidKV	+0.231	[+0.191, +0.272]	400

Table 20: 95% CIs for the remaining headline deltas. Cross-architecture rows (Table 5) use the four-budget grid and the same cluster bootstrap; 16K rows are the second half of Table 9; the DBTrimKV row is Table 16 ($N = 120$ inputs, budgets {128, 256, 512}). The Mistral NIAH-MK3 showcase entries are Wilson score intervals on $N = 100$. The only headline delta whose interval includes zero is Qwen2.5-3B at 16K — the gate-misfire cell that Section 4.2 already reports as a shrunken effect.

Quantity	Point	95% CI	N
Yi-1.5-9B Δ (SnapKV, 4-budget)	+0.390	[+0.338, +0.444]	200
Llama-3.1-8B Δ (SnapKV, 4-budget)	+0.109	[+0.069, +0.154]	200
Qwen2.5-1.5B 16K Δ	+0.032	[+0.013, +0.053]	200
Qwen2.5-3B 16K Δ	+0.034	[−0.002, +0.071]	200
Qwen2.5-14B 16K Δ	+0.071	[+0.034, +0.107]	120
Mistral-7B 16K Δ	+0.116	[+0.079, +0.156]	200
DBTrimKV wrapper Δ (RULER 4K)	+0.233	[+0.172, +0.297]	120
Mistral 4K NIAH-MK3, gated acc. @ $b=0.0625$	0.89	[0.814, 0.937]	100
Mistral 4K NIAH-MK3, plain acc. @ $b=0.0625$	0.00	[0.000, 0.037]	100

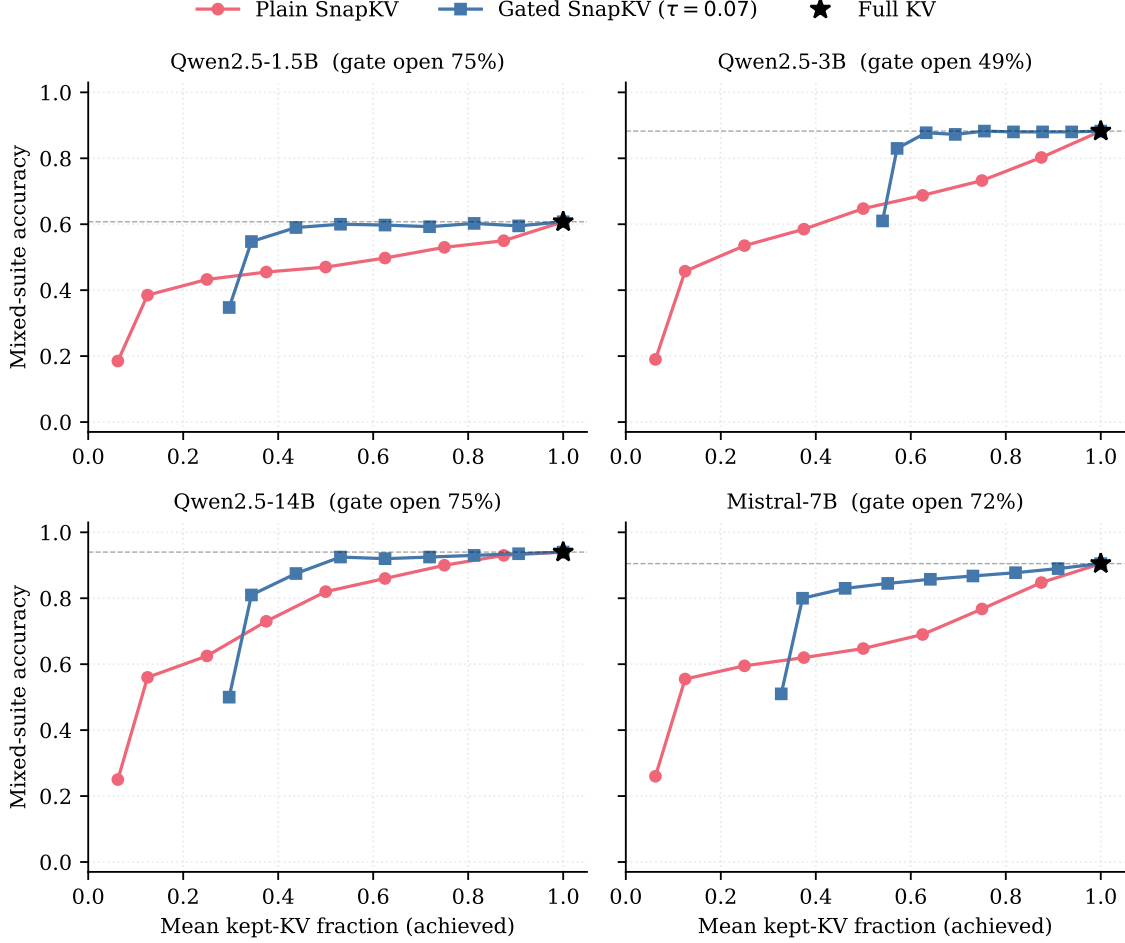


Figure 3: Accuracy vs. *achieved* kept-KV fraction (mean over inputs of n_{kept}/T) on the mixed 4K suite (NIAH-MK3 + VT + FWE + QA_1), SnapKV. Because the gate falls back to the full cache when it closes, gated points sit at larger kept-KV than their nominal budget; the honest comparison is at matched achieved cache size. For every target kept-KV fraction $\gtrsim 1/3$ (Qwen3B: $\gtrsim 0.57$) the gated curve is the Pareto frontier, by up to +19pp over plain at equal cache size; at the most aggressive nominal budget ($b = 0.0625$, leftmost gated point) the gated point is dominated by plain eviction at a matched cache size, and $\tau = 0.07$ caps achievable compression at 1 – open-fraction of the cache (0.25–0.51 across models). All gated results use the post-hoc $\tau = 0.07$ reconstruction (Appendix G); the reconstruction reproduces the stored gated outcomes exactly on the three runs executed at $\tau = 0.07$.

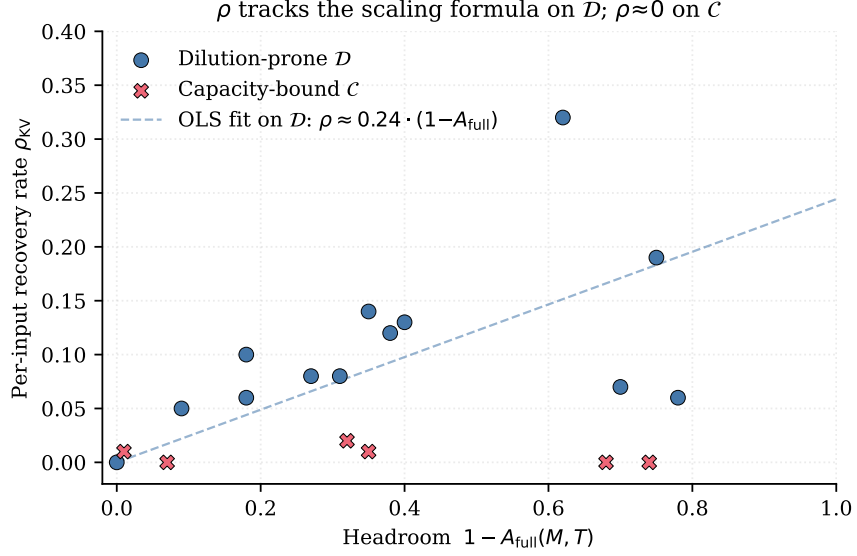


Figure 4: Per-input recovery rate ρ vs. accuracy headroom $1 - A_{full}$ across the $(model, context, task)$ cells of Table 13. Dilution-prone tasks (filled circles) lie on a band of slope ≈ 0.4 ; NIAH-MK3 (open triangles) lies on the $\rho = 0$ axis regardless of headroom. The two populations are visibly separated.

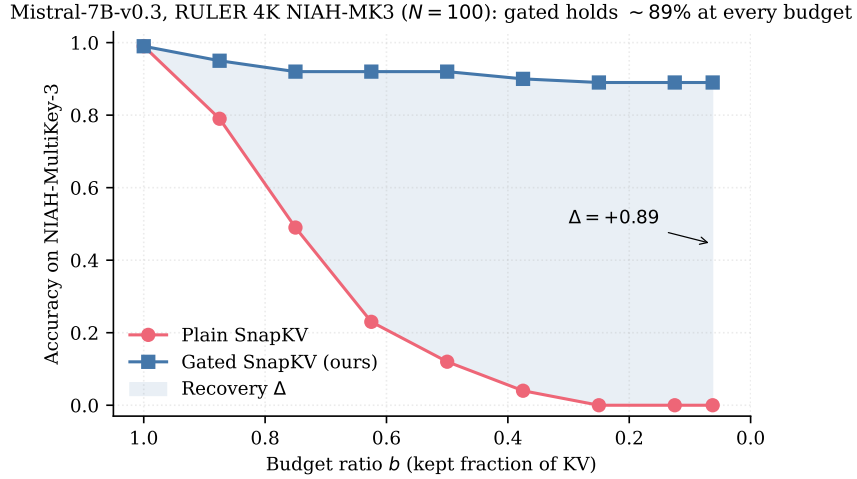


Figure 5: Mistral-7B 4K NIAH-MK3, $N = 100$. Accuracy vs. KV budget for plain SnapKV (solid, collapses to 0) and gated SnapKV (dashed, 0.89 flat across budgets). The gate prevents the catastrophic failure mode without retraining the base evictor or seeing labels.

Per-input recovery distribution by task (pooled across model×context cells)

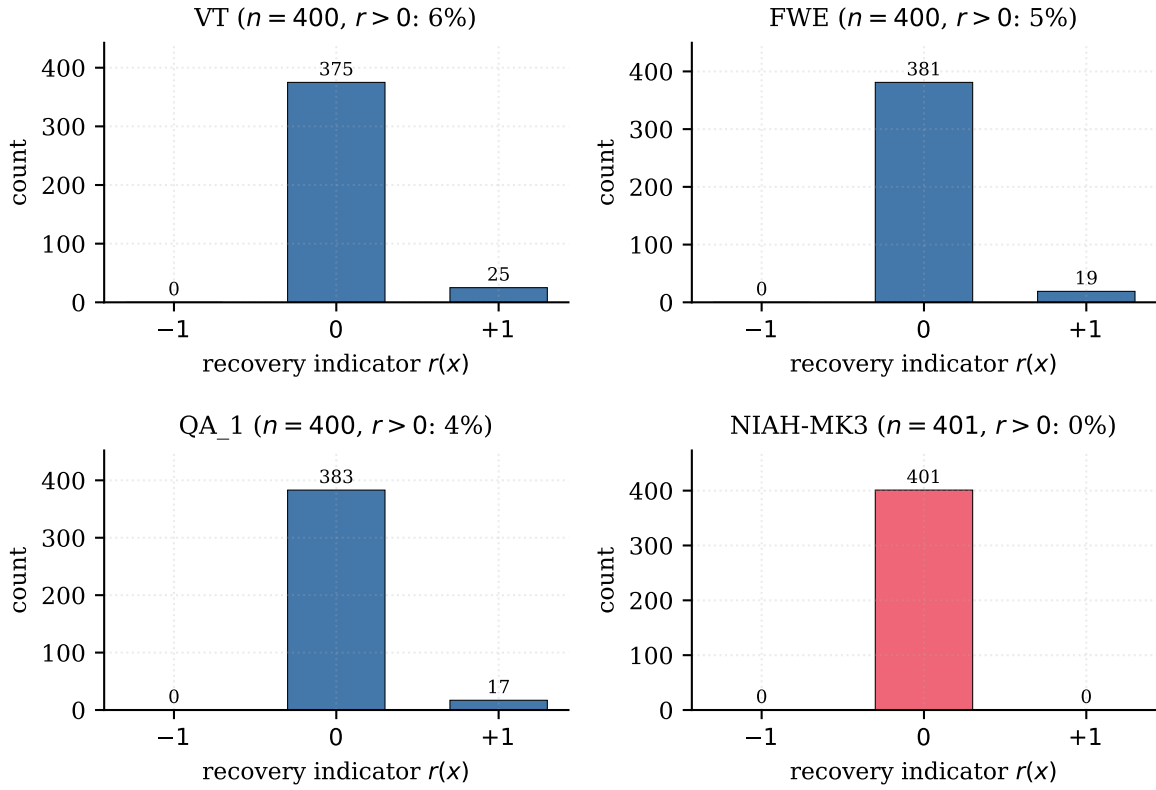


Figure 6: Per-input recovery indicator $r(x)$ pooled across model and context cells, faceted by RULER task. The three dilution-prone tasks show a non-trivial right tail at $r=+1$ with zero mass at $r=-1$; NIAH-MK3 is concentrated entirely at $r=0$, consistent with the capacity-bound regime.

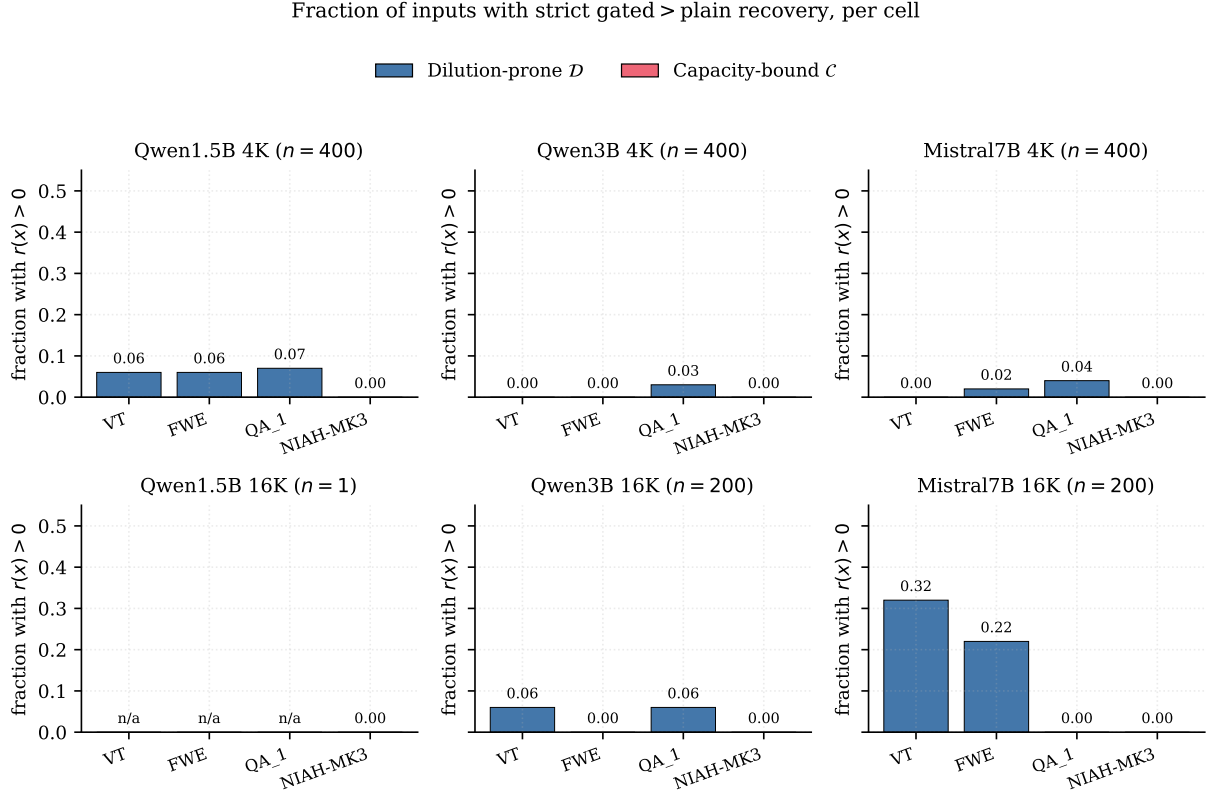


Figure 7: Fraction of inputs with strict recovery $r(x) > 0$ per task, faceted by (model, context) cell. Dilution-prone bars are non-trivial in every cell with data, while NIAH-MK3 stays at or near zero, visualising the partition input by input.